Investments in Human Capital in the Age of AI

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Abstract

How does Artificial Intelligence reshape human capital investment decisions? I develop a task-based model where risk-averse agents make irreversible training choices based on AI's expected impact and uncertainty. Calibrated to the US labor market, the framework decomposes AI shocks into expected automation, expected augmentation, and their associated uncertainties. The model yields three key results. First, expected automation drives a structural reallocation, causing young students to systematically avoid training paths facing high automation risk, while choosing paths requiring more manual or social skills. Second, average expected wages rise, but their distribution depends on the responsiveness of training choices made by the new generation: when it is high, labor scarcity offsets the direct impact of AI and wage gains are nearly uniform across training paths; when it is low, labor reallocation is weaker, and wage gains are more concentrated in the safest paths. Third, automation uncertainty generates an endogenous risk premium.

Keywords: artificial intelligence, human capital, training choices, automation, augmentation, uncertainty, task-based model.
JEL codes: D81, I26, J24, J31, O33.

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1 Introduction

For most of the twentieth century, technological progress pointed in the direction of incentivizing investments in human capital. Successive waves of automation primarily displaced routine, mechanical tasks concentrated in low- and middle-skill occupations. This process raised the demand for high-skill workers with tertiary education (Acemoglu and Autor, 2011; Autor et al., 2003). The secular rise in the college wage premium and in educational attainment throughout the last 50 years is largely a product of this dynamic (Burstein et al., 2019; Acemoglu and Restrepo, 2022; Autor, 2019).

Artificial Intelligence (AI) could disrupt this long-standing relationship. Unlike previous technologies, generative AI demonstrates a high degree of proficiency precisely in the non-routine, cognitive tasks that have historically been the exclusive domain of highly educated workers (Eloundou et al., 2024; Tomlinson et al., 2025; Massenkoff and McCrory, 2026; Webb, 2019). The implications are potentially far-reaching: the very occupations that attracted generations of students to invest heavily in human capital are now among the most exposed to this new technology.

AI's impact on training choices also has important implications for expected wages. If individuals systematically avoid training paths that are highly exposed to AI, this creates a scarcity of workers with that training, which mechanically pushes up their expected wages. Conversely, the inflow of workers into safer training paths creates congestion that puts downward pressure on wages in those training paths. In this paper, I show that this mechanism has the potential to counteract the direct effects of AI: training paths that are expected to be negatively impacted by AI are chosen less by the new generation of workers, and this scarcity partially offsets the direct negative impact of AI on wages, while training paths that are expected to be positively impacted by AI attract workers, and this congestion partially offsets the direct positive impact of AI on wages. How far this mechanism goes depends on how responsive training choices are to expected wages. When they are highly responsive, the two forces balance each other, and expected wages increase by about +15% in all training paths. When they are less responsive, fewer workers reallocate, and the impact of AI shows up in wages rather than in employment.

Another critical feature of this technological breakthrough, however, is its profound uncertainty. Not only is the net effect of AI on labor demand unknown, but its expected impact and the magnitude of the uncertainty also vary across occupations and training paths. It is very difficult to predict exactly which tasks will be automated and which will be augmented. Hence, risk-averse individuals may systematically avoid training paths that are relatively uncertain, irrespective of their expected returns. This flight from uncertainty operates through the same scarcity channel: it reduces labor supply

in risky paths and generates an endogenous risk premium that pushes their expected wages upward.

To formalize and quantify these dynamics, I develop a task-based model in which heterogeneous, risk-averse agents choose the optimal irreversible human capital investment among 13 training paths to maximize their expected lifetime utility, net of idiosyncratic investment costs. The key innovation is the decomposition of the AI shock for each training path j into four distinct components: the expected automation share $\mu_{A,j}$, the expected augmentation index $\mu_{B,j}$, and their respective variances $\sigma^2_{A,j}$ and $\sigma^2_{B,j}$, which capture the uncertainty about AI's ultimate capabilities. The individual discrete choice problem aggregates to a tractable multinomial logit equilibrium.

I calibrate the model to the US labor market. The task-level AI shocks are derived from Althoff and Reichardt (2026), who computed for each task whether it is theoretically automatable with generative AI, as well as the potential augmentation factor if generative AI instead acts as an augmentation technology in that task. Moreover, I rely on the current AI task penetration (Massenkoff and McCrory, 2026), which measures AI usage in each task according to recent Anthropic data. The employment shares across the 13 training paths are constructed from the IPUMS American Community Survey (ACS) for 2022 (the pre-AI baseline year) and from the employment data of the Bureau of Labor Statistics (BLS), while the median wages are collected from the BLS. I recover the structural training cost parameters by inverting the pre-AI steady-state equilibrium (2022), and simulate the resulting post-AI equilibrium.

A key parameter is the responsiveness of training choices to expected wages, governed by the dispersion of idiosyncratic preferences. Existing estimates of this responsiveness are diverse and are not directly applicable to the AI context, so I do not take a stand on a single value. Instead, I consider three scenarios, each disciplined by a different strand of evidence: *high responsiveness*, from the occupational choice model of Althoff and Reichardt (2026); *medium responsiveness*, from Roy models of occupational and sectoral choice (Burstein et al. (2019), Hsieh et al. (2019), Galle et al. (2023)); and *low responsiveness*, from the evidence on college major choice (Long et al., 2015). I treat the three scenarios as equally plausible.

The model delivers three main results, linking theoretical mechanisms to their quantitative implications for the US economy. First, expected automation is the primary driver of labor reallocation, shifting workers away from highly exposed training paths (legal professions, computer science, and business) toward safer alternatives (technical manual jobs and education) or to no training at all. For example, computer science loses between 4% and 11% of its workforce, depending on the responsiveness of training choices, while the ranking of training paths is the same in all scenarios. Also, uncertainty about automation amplifies this phenomenon: part of the reallocation is driven

by pure risk aversion.

Second, AI raises average expected wages by about +15% to +16% in all scenarios, but the responsiveness of training choices determines how these gains are distributed across training paths. When training choices are highly responsive, the scarcity of workers in the most exposed training paths pushes up their expected wages and nearly offsets the direct negative effect of AI, so that expected wages increase by about +15% in all training paths. When training choices are less responsive, the scarcity channel is weaker, and wage gains range from about +10% in the most exposed paths to about +21% in the safest one.

Third, uncertainty generates an endogenous risk premium (up to +2.2%, +1.2%, and +0.7% under high, medium, and low responsiveness, respectively). This premium is a compensating differential for risk in the spirit of Rosen (1986): risk-averse agents avoid uncertain paths, and the resulting labor scarcity raises expected wages in those paths. What is new is not the mechanism itself, but the source and structure of the risk: it stems from uncertainty about AI's ultimate capabilities, it differs across training paths, and it is priced when training choices are made and are irreversible.

These three mechanisms interact in ways that are difficult to detect using reduced-form methods alone. A highly exposed training path may not see its expected wage fall, thanks to labor scarcity, even if AI has a negative direct expected impact. Conversely, a safer path may experience disappointing wage growth due to labor market congestion. This work provides a structural framework to disentangle these intertwined effects.

This paper contributes to three strands of the literature. First, it builds on the large empirical literature on AI's impact on the labor market (among many others, Acemoglu and Restrepo 2018; Autor and Thompson 2025; Agrawal et al. 2025; Ide and Talamàs 2025; Acemoglu et al. 2025; Gans and Goldfarb 2026), by providing a structural model that allows for a clean decomposition of the supply-side response to the future expected demand shift and uncertainty related to AI. Second, it contributes to the literature on human capital investment under uncertainty (Rosen, 1986; Levhari and Weiss, 1974; Cunha et al., 2005; Cunha and Heckman, 2016), by introducing a new form of risk in the context of human capital investment decisions, and by showing how AI-related uncertainty that differs across training paths is priced in equilibrium. Third, it connects to the macro literature on skill-biased technological change (Acemoglu and Autor, 2011; Autor et al., 2003; Autor, 2019) by characterizing the novel mechanism through which AI reshapes economic incentives in ways that are qualitatively different from prior automation waves.

The closest paper to this work is Althoff and Reichardt (2026), who provide the foundational task-level framework for how AI simultaneously automates, augments, and simplifies human labor. While they embed these mechanisms into a rich dynamic

model of occupational choice, they treat initial skills as exogenous and focus on skill accumulation on the job. Furthermore, their analysis evaluates the labor market transition under the assumption that uncertainty is symmetric across occupations. This paper takes a complementary approach by explicitly focusing on the endogenous pre-labor market training choice and modeling the training-specific uncertainty surrounding AI's ultimate capabilities.

The rest of the paper is organized as follows. Section 2 develops the theoretical model. Section 3 describes the measurement of the AI shock. Section 4 describes the calibration strategy, and Section 5 presents the quantitative simulation results. Section 6 presents a sensitivity analysis using alternative parameterizations. Finally, Section 7 concludes.

2 The Model

I consider an economy where agents live for two periods: young ($g = y$) and old ($g = o$). The economy features a finite set of potential training paths. Each training path involves a period of training (when young) followed by a working period (when old). The only exception is the case of no training, which represents a fallback option with no investment in human capital. Training paths may be of two types: a *technical* training path (e.g., plumber, cook) is characterized by a period of apprenticeship as training, while a *cognitive* training path involves a period of studying in college in a certain field before entering the labor market. Let us denote the different training paths by $j \in \{0, 1, ..., J\}$ ($j = 0$ refers to the case of no training). When young ($g = y$), agents are endowed with a unit of time and choose a training path j. This choice involves an irreversible investment in human capital with uncertain returns. In the working period ($g = o$), agents supply labor inelastically, realize their wage (which depends on the chosen training and labor market conditions), and consume. To better understand the timing of the model, Figure 1 provides a visual representation of the sequence of events. Notice that at each point in time, there exist two generations of workers: the young generation is making training choices, without yet entering the labor market, while the old generation is employed and supplying their (trained) labor inelastically. Workers who were in their training path before AI had to choose without anticipating any AI shocks. In the empirical part, I will use the workers' distribution across training paths in 2022 as the pre-AI steady state, and I will compare outcomes between the pre-AI and post-AI working periods. After the introduction of AI, the new generation of workers (ideally students finishing high school after 2022) takes into account AI shocks when choosing their optimal training path. After they choose the optimal training, they enter the labor market where they supply their (trained) labor inelastically.

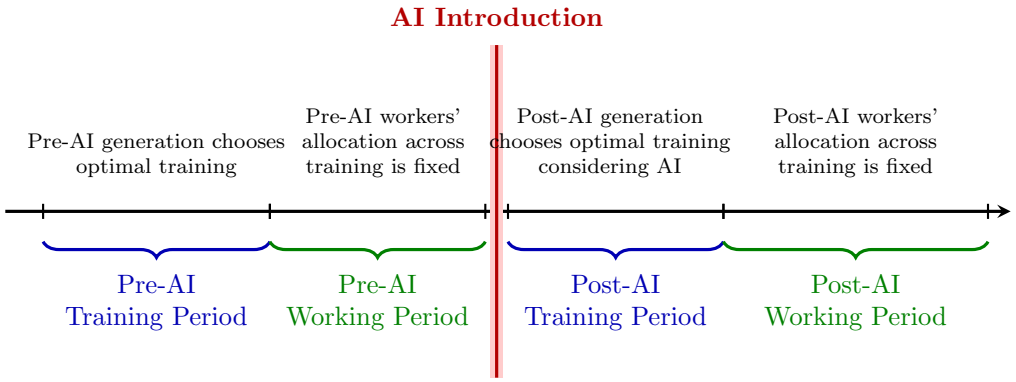


Figure 1: Timeline of the Model.

Notes: The vertical red bar marks the introduction of AI. The pre-AI generation (left) makes training choices and works without anticipating AI shocks. The post-AI generation (right) observes the expected AI shocks ($\mu_{A,j}$, $\mu_{B,j}$) and their uncertainty ($\sigma^2_{A,j}$, $\sigma^2_{B,j}$) before making their irreversible training choice.

2.1 Working Period

A representative firm demands labor for each training path. The output per worker for training j is

$$Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} y_\tau^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \tag{1}$$

where $\mathcal{T}_j$ is the set of relevant tasks for training j, y_τ is the output of task τ, and $\theta_{j,\tau}$ is the importance weight of task τ in the production for training j.
The production function for task τ is

$$y_\tau(l_\tau, k_\tau) = \begin{cases} l_\tau \gamma_\tau & \text{if } \tau \in \mathcal{N}_j \\ l_\tau \gamma_\tau + \phi_\tau k_\tau, & \text{if } \tau \in \mathcal{A}_j \end{cases}$$

where $\mathcal{A}_j$ and $\mathcal{N}_j$ are, respectively, the sets of automatable and non-automatable tasks in training j, l_τ is the share of time a worker with training j spends in task τ, and k_τ is the capital devoted to task τ. Finally, γ_τ and ϕ_τ are parameters that capture the productivity of labor and capital in task τ. In this framework, automation refers to an expansion of the set $\mathcal{A}_j$, and labor augmentation refers to an increase in γ_τ. Thus, in the rest of the paper, I will refer to AI automation when AI can entirely perform tasks that were previously done by humans (i.e., when $\mathcal{A}_j$ expands); on the other hand, I will refer to AI augmentation when AI increases the productivity of human workers without replacing them entirely at the task level (i.e., when γ_τ increases, but $\mathcal{A}_j$ does not expand).
The representative firm takes the number of workers with a certain training, the wage,

and the rental rate of capital as given. Formally, the problem solved by the representative firm is

$$\begin{aligned}
\max_{l_{j,\tau},k_{j,\tau}} L_j \left(p_j Y_j - w_j - R \sum_{\tau \in \mathcal{A}_j} k_{j,\tau} \right) \\
\text{subject to } Y_j = \left(\sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}^{\frac{1}{\rho}} \tilde{y}_\tau^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \\
\tilde{y}_\tau = y_\tau(l_{j,\tau}, k_{j,\tau}) \\
\sum_{\tau \in \mathcal{T}_j} l_{j,\tau} = 1
\end{aligned}$$

where L_j is the share of workers with training j in the labor market. Importantly, L_j is fixed from the previous training period choice and cannot be adjusted until the next period (i.e., the next generation).

In what follows, I assume that, for automatable tasks in $\mathcal{A}_j$, the unit cost of producing a task with capital is lower than the cost of producing it with labor (i.e., $\frac{R}{\phi_\tau} < \frac{w_j}{\gamma_\tau}$ for all $\tau \in \mathcal{A}_j$). It follows that all automatable tasks are produced with capital, and the labor demand for task τ is zero for $\tau \in \mathcal{A}_j$. Formally,

$$\begin{cases} y_\tau = l_\tau \gamma_\tau, & \text{if } \tau \in \mathcal{N}_j \\ y_\tau = \phi_\tau k_\tau, & \text{if } \tau \in \mathcal{A}_j \end{cases} \tag{2}$$

By solving the maximization problem, the optimal time allocation of labor is

$$\begin{cases} l_{j,\tau} = \frac{\theta_{j,\tau} \gamma_\tau^{\rho-1}}{\sum_{\tau' \in \mathcal{N}_j} \theta_{j,\tau'} \gamma_{\tau'}^{\rho-1}} & \text{if } \tau \in \mathcal{N}_j \\ l_{j,\tau} = 0 & \text{if } \tau \in \mathcal{A}_j \end{cases} \tag{3}$$

and the optimal capital allocation for each training-task pair is

$$\begin{cases} k_{j,\tau} = 0 & \text{if } \tau \in \mathcal{N}_j \\ k_{j,\tau} = \theta_{j,\tau} \phi_\tau^{\rho-1} Y_j \left(\frac{p_j}{R} \right)^\rho & \text{if } \tau \in \mathcal{A}_j \end{cases} \tag{4}$$

Intuitively, the optimal allocation of labor shows that workers spend more time in tasks that are more important. Similarly, more capital is allocated to tasks that weigh more. Also, more capital is allocated to training paths that are more productive (i.e., with higher Y_j).

By plugging (3) and (4) into (1), the optimal production per worker in training j is

$$Y_j = \Gamma_j^{\frac{\rho}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}} \tag{5}$$

where $\Gamma_j = 1 - \sum_{\tau \in \mathcal{A}_j} \theta_{j,\tau} \left(\frac{R/\phi_\tau}{p_j} \right)^{1-\rho}$ is the labor share for training j.

Finally, since markets are perfectly competitive, the wage is given by

$$w_j = p_j Y_j \Gamma_j = p_j \Gamma_j^{\frac{1}{1-\rho}} \left(\sum_{\tau \in \mathcal{N}_j} \theta_{j,\tau} \gamma_\tau^{\rho-1} \right)^{\frac{1}{\rho-1}} \tag{6}$$

Notice that if all tasks are automated ($\mathcal{N}_j = \emptyset$), labor has no use and $w_j = 0$. As long as some tasks remain performed by humans, however, the wage depends on the output per worker, the labor share, and the price (determined in general equilibrium) of what is produced with a given training. All these variables depend on the AI capabilities.

2.2 Training Period

Before entering the working period, agents must choose their training (or forgo training altogether). Agents are risk-averse, with preferences described by a CRRA utility function $u(w) = \frac{w^{1-\eta}}{1-\eta}$. Individuals are heterogeneous with respect to a vector of net costs $C_i = [C_{i0}, C_{i1}, \ldots, C_{iJ}]$, which aggregates all non-pecuniary disutility and direct expenses specific to each training path. For example, if a worker strongly dislikes math, the cost of choosing engineering will be very high, whereas if she particularly enjoys literature, the cost of choosing humanities will be very low.

A young agent i chooses the training path j that maximizes the discounted expected utility of future earnings, net of the investment cost:

$$V_i^y(C_i) = \max_j \{\beta \mathbb{E}[u(w_j)] - C_{ij}\} \tag{7}$$

where $\beta \in (0,1)$ is the discount factor, the expectation is taken over the AI shocks realized in the working period, and the wage for training path j is given by the competitive wage equation (6). As in other standard task-based models, automation has an ambiguous impact because it shrinks the set of tasks performed by humans (displacement effect), but it increases productivity in the remaining tasks due to the cost savings in the automated tasks (productivity effect); also, it generates a change in the equilibrium prices, which has a direct impact on wages. Similarly, augmentation has an ambiguous impact on wages because it increases the productivity of labor, but it

changes the equilibrium prices. However, when young, agents do not know what the final set of automatable tasks $\mathcal{A}_j$ in each training j will be, nor which tasks will be augmented.

To formalize the uncertainty, I build on the task-level structure that will be explained in more detail in Section 3. As far as this section is concerned, let $a_\tau \in \{0,1\}$ be a Bernoulli random variable indicating whether task τ will eventually be automated (i.e., when young workers will be in their working period), and let $b_\tau \in \{0,1\}$ be a Bernoulli random variable indicating whether it will be augmented. Following the classification in Section 3, the tasks of training path j can be partitioned into three disjoint sets: tasks that are already automated (during the post-AI training period), $\mathcal{S}_{A,j}^{cur}$ (for which $a_\tau = 1$ with certainty); tasks that are potentially automatable but not yet automated, $\mathcal{S}_{A,j}^{pot}$ (for which a_τ is random); and tasks that are not automatable by AI, $\mathcal{S}_j^{non}$ (for which $a_\tau = 0$ with certainty). I denote by $S_{A,j}^{cur}$, $S_{A,j}^{pot}$ and S_j^{non} the corresponding weighted shares, so that $S_{A,j}^{cur} + S_{A,j}^{pot} + S_j^{non} = 1$.

The aggregate automation share for training path j is

$$A_j = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}\, a_\tau \ \in [0,1]$$

Notice that while $\mathcal{A}_j$ is a set of tasks that are automated, A_j is a continuous variable representing the weighted share of automated tasks ($A_j = 0$ if no tasks are automated and $A_j = 1$ if all tasks are automated).
The aggregate augmentation factor (the sum in the last term of Equation (6)) is

$$\Pi_j = \sum_{\tau \in \mathcal{T}_j} \theta_{j,\tau}(1 - a_\tau)\gamma_\tau^{\rho-1}$$

where γ_τ is the realized labor productivity in task τ ($\gamma_\tau = 1$ if not augmented). Notice that A_j enters Π_j directly through the terms $(1 - a_\tau)$: automating a task removes it from the bundle of tasks performed by humans. Hence, A_j and Π_j are mechanically correlated even when automation and augmentation shocks are unrelated. To keep the model tractable, I impose the following assumption.

Assumption 1 (Separable AI Shocks)**.**

(i) Tasks that are theoretically automatable by AI face only automation risk: for $\tau \in \mathcal{S}_{A,j}^{cur} \cup \mathcal{S}_{A,j}^{pot}$, $\gamma_\tau = 1$ whenever the task is not automated. Augmentation can only occur in non-automatable tasks, for which $\gamma_\tau = 1 + b_\tau(g_\tau - 1)$, where $g_\tau > 1$ is the potential augmentation factor of task τ if the task is actually augmented ($g_\tau = 1.1$ means a 10% increase in productivity).

(ii) Automation shocks $\{a_\tau\}$ and augmentation shocks $\{b_\tau\}$ are independent.

Under Assumption 1, the aggregate augmentation factor becomes

$$\Pi_j = \underbrace{\sum_{\tau \in \mathcal{S}_{A,j}^{pot}} \theta_{j,\tau}(1-a_\tau)}_{=S_{A,j}^{pot}-(A_j-S_{A,j}^{cur})} + \underbrace{\sum_{\tau \in \mathcal{S}_j^{non}} \theta_{j,\tau}\,\gamma_\tau^{\rho-1}}_{\equiv B_j} = \left(1 - S_j^{non}\right) - A_j + B_j \tag{8}$$

Thus, Π_j is linear in two independent aggregate shocks: the automation share A_j, which depends only on the a_τ's, and the augmentation index B_j, which depends only on the b_τ's. When $\rho < 1$ (as is the case in the calibrated model), we have $g_\tau^{\rho-1} < 1$, so that a *lower* B_j corresponds to *more* augmentation. Let $\mu_{A,j} \equiv \mathbb{E}[A_j]$ and $\sigma_{A,j}^2 \equiv Var(A_j)$ denote the mean and variance of the automation share, and let $\mu_{B,j} \equiv \mathbb{E}[B_j]$ and $\sigma_{B,j}^2 \equiv Var(B_j)$ denote the corresponding moments of the augmentation index. By independence, $Cov(A_j, B_j) = 0$, and the expected augmentation factor is

$$\mu_{\Pi,j} \equiv \mathbb{E}[\Pi_j] = 1 - S_j^{non} - \mu_{A,j} + \mu_{B,j}.$$

Because b_τ is Bernoulli, $\gamma_\tau^{\rho-1}$ takes only the values 1 and $g_\tau^{\rho-1}$, so the moments of B_j are exact. The explicit construction of these moments from the task-level primitives is detailed in Section 3.

Assumption 1 comes at a cost. First, for tasks that are automatable but not yet automated, part (i) rules out augmentation even when it is currently observed in the data. While this is a simplifying assumption, it is not entirely unrealistic. When workers are in their training period, they are concerned with the potential of AI in the future. As a result, it is reasonable to assume that they focus on the risk of automation even if the task is currently augmented. Second, part (ii) rules out that the "success of AI" is correlated across the two types of shocks. While there could be arguments for such a correlation, it is not implausible that some training paths will experience more automation without corresponding augmentation and vice versa. Nonetheless, developing a more general model that allows for correlation between automation and augmentation shocks would be an important extension for future work.

Importantly, all four parameters $(\mu_{A,j}, \sigma_{A,j}^2, \mu_{B,j}, \sigma_{B,j}^2)$ are training-specific. Thus, there are some training paths characterized by higher risk of automation and others where augmentation is more likely; also, some training paths could be relatively less subject to both automation and augmentation, but they can still be indirectly affected through demand spillovers. Moreover, the variance is also training-specific, and it captures the fact that some training paths are potentially more uncertain than others.

Notice that expected utility can be expressed as

$$\mathbb{E}[u(w_j)] = \frac{1}{1-\eta}\mathbb{E}[w_j^{1-\eta}] = \frac{1}{1-\eta}\mathbb{E}\left[\exp\left((1-\eta)\ln w_j\right)\right]$$

When the AI shocks are realized, the labor allocation $\vec{L}$ has already been fixed by the training choices of the previous period. Therefore, the response of prices to the realized shocks is a static market-clearing adjustment. I assume that agents form rational expectations over this adjustment: they anticipate how the equilibrium price p_j responds to the realization of A_j and B_j, while the level of the price is evaluated at the expected shocks, $\ln p_j(\vec{L}, \vec{\mu}_A, \vec{\mu}_B)$, consistently with the first-order approximation below.

Using Equation (8), the log-wage can be written as a function of the two independent shocks (A_j, B_j):

$$\ln w_j = \ln p_j + \ln Y_j + \ln \Gamma_j, \qquad \ln Y_j = \frac{\rho}{1-\rho}\ln \Gamma_j(A_j) + \frac{1}{\rho-1}\ln\left(1 - S_j^{non} - A_j + B_j\right) \tag{9}$$

where, following Althoff and Reichardt (2026), I hold fixed the cost of producing automatable tasks with capital relative to labor, χ (see Sections 2.3 and 4), so that the labor share Γ_j depends on A_j but not on p_j. With CES demand across occupational goods (elasticity ϵ) and the aggregate price index as numéraire, market clearing with L_j fixed implies

$$d\ln p_j = -\frac{1-s_j}{\epsilon}\, d\ln Y_j + \sum_{k\neq j}\frac{s_k}{\epsilon}\, d\ln Y_k \tag{10}$$

where s_j is the expenditure share of good j. An increase in output per worker in path j lowers its price, and the factor $(1-s_j)$ accounts for the weight of path j in the aggregate index. In the quantitative analysis, I calibrate s_j with the 2022 wage shares (Section 4), although the linearization is evaluated at the expected post-AI equilibrium; this is accurate as long as expenditure shares do not change much between the two equilibria.

Because $\ln w_j$ is a non-linear function of A_j and B_j, its exact distribution is analytically intractable. To restore tractability, I apply a first-order Taylor expansion (the Delta method) to $\ln w_j$ around the expected shocks $(\mu_{A,j}, \mu_{B,j})$. The linearized log-wage is

$$\ln w_j \approx \ln w_j\big|_{\mu_{A,j},\mu_{B,j}} + \left.\frac{\partial \ln w_j}{\partial A_j}\right|_{\mu_{A,j},\mu_{B,j}}(A_j - \mu_{A,j}) + \left.\frac{\partial \ln w_j}{\partial B_j}\right|_{\mu_{A,j},\mu_{B,j}}(B_j - \mu_{B,j})$$

Combining Equations (9) and (10), and defining the pass-through of output to wages net of the price response as

$$\lambda_j \equiv 1 - \frac{1-s_j}{\epsilon},$$

which lies in $(0,1)$ as long as $\epsilon > 1 - s_j$ (a condition satisfied by the calibrated value $\epsilon = 1.57$), the sensitivities, evaluated at the expected shocks, are

$$\frac{\partial \ln w_j}{\partial A_j} = \underbrace{\left[1 + \lambda_j \frac{\rho}{1-\rho}\right] \frac{\partial \ln \Gamma_j}{\partial A_j}}_{\text{displacement } (<0)} + \underbrace{\frac{\lambda_j}{1-\rho} \frac{1}{\mu_{\Pi,j}}}_{\text{productivity } (>0)}, \qquad \frac{\partial \ln w_j}{\partial B_j} = -\frac{\lambda_j}{1-\rho} \frac{1}{\mu_{\Pi,j}} < 0 \quad (11)$$

The sensitivity to automation is a *total* derivative: an increase in A_j lowers the labor share Γ_j (displacement effect), but it also removes tasks from the human bundle, lowering Π_j and raising output per worker (productivity effect). Part of the productivity gain, however, is dissipated by the fall in the price of good j, which is why the productivity effect is scaled by $\lambda_j < 1$. The sensitivity to B_j is negative, because a lower B_j means more augmentation and hence higher wages; its magnitude is also dampened by the price response.

Finally, Equation (10) implies that the wage in path j also depends on the shocks of the other paths $k \neq j$, through the aggregate price index, with weight s_k/ϵ. Since shocks are independent across training paths, these cross-path terms would add $\sum_{k \neq j} \left(\frac{s_k}{\epsilon} \frac{\partial \ln Y_k}{\partial A_k}\right)^2 \sigma^2_{A,k}$ and the analogous terms in $\sigma^2_{B,k}$ to the variance of $\ln w_j$. I abstract from these terms in what follows. This omission is less restrictive than it may seem: the sum $\sum_{k \neq j}$ is almost identical across training paths, since it differs only by the path's own term. A term common to all the possible alternatives cancels out in the logit choice (Equation (18)), so the error in the allocation of workers is only of order $(s_j/\epsilon)^2$ times the path's own variance.

Under this linear approximation, the first two moments of $\ln w_j$ are well-defined. The expected future log-wage is

$$\mathbb{E}[\ln w_j] = \ln p_j(\{L_k, \mu_{A,k}, \mu_{B,k}\}_{k=0}^J) + \frac{1}{1-\rho} \ln \Gamma_j(\mu_{A,j}) + \frac{1}{\rho - 1} \ln \mu_{\Pi,j}$$

where $\mu_{\Pi,j} = 1 - S_j^{non} - \mu_{A,j} + \mu_{B,j}$; the endogenous expected price depends on the reallocation of labor supply and on expected automation and augmentation; the expected labor share in each training j depends on expected automation; finally, the last term captures the expected productivity, which depends on expected automation and on expected augmentation of the non-automatable tasks. Since A_j and B_j are independent, the overall variance is the sum of the variances of the underlying shocks, weighted by the squared sensitivities:

$$Var(\ln w_j) = \left(\frac{\partial \ln w_j}{\partial A_j}\bigg|_{\mu_{A,j}, \mu_{B,j}} \right)^2 \sigma^2_{A,j} + \left(\frac{\partial \ln w_j}{\partial B_j}\bigg|_{\mu_{A,j}, \mu_{B,j}} \right)^2 \sigma^2_{B,j} \quad (12)$$

To compute expected utility, I further assume that $\ln w_j$ is normally distributed, so that wages are log-normal. Under this assumption, expected utility can be computed using the moment-generating function of the normal distribution applied to $\ln w_j$:

$$\mathbb{E}[u(w_j)] = \frac{1}{1-\eta} \exp\left((1-\eta)\mathbb{E}(\ln w_j) + \frac{(1-\eta)^2}{2} Var(\ln w_j)\right) \tag{13}$$

Let us introduce the certainty equivalent $w_{CE,j}$, which is the guaranteed wage that yields the same expected utility, $u(w_{CE,j}) = \mathbb{E}[u(w_j)]$. Plugging in Equation (13) and solving for $w_{CE,j}$, we obtain

$$w_{CE,j} = \exp\left(\mathbb{E}(\ln w_j) - \frac{\eta-1}{2} Var(\ln w_j)\right) \tag{14}$$

Instead of working with the certainty equivalent wage, it is convenient to work with its logarithm, which I denote by Ω_j:

$$\Omega_j = \ln w_{CE,j} = \mathbb{E}(\ln w_j) - \frac{\eta-1}{2} Var(\ln w_j) \tag{15}$$

Throughout, I assume $\eta > 1$, which is what makes the term $\frac{\eta-1}{2} Var(\ln w_j)$ a penalty: uncertainty about log-wages lowers the certainty equivalent. Thus, the final expression for the certainty equivalent Ω_j is

$$\Omega_j = \underbrace{\ln p_j + \frac{1}{1-\rho} \ln \Gamma_j + \frac{1}{\rho-1} \ln \mu_{\Pi,j}}_{\mathbb{E}(\ln w_j)} - \underbrace{\frac{\eta-1}{2} Var(\ln w_j)}_{\text{Risk premium}} \tag{16}$$

which formalizes exactly how agents penalize uncertainty depending on their degree of risk aversion η. Importantly, this model creates a training-specific risk premium that is linked to the relative uncertainty regarding future automation and augmentation due to AI in each training path.

Substituting $\mathbb{E}[u(w_j)]$ with $u(w_{CE,j})$ in Equation (7), each training path is fully summarized, from the agent's perspective, by its log certainty equivalent Ω_j. To keep the discrete choice problem tractable, I assume that agents rank training paths according to the log certainty equivalent, net of a (normalized) training cost C^*_{ij}:

$$\max_j \left\{\beta \Omega_j - C^*_{ij}\right\} \tag{17}$$

Risk aversion η governs the valuation of uncertainty *within* a training path, through the risk premium in Ω_j (Equation (16)), while the choice *across* paths is linear in Ω_j.

I assume that the normalized cost is determined by a training-specific common

average cost, $\bar{C}_j^*$, and an idiosyncratic preference component, such that $C_{ij}^* = \bar{C}_j^* - \xi_{ij}$. Assuming the idiosyncratic shocks ξ_{ij} are independently and identically distributed following a Type I Extreme Value (Gumbel) distribution with location parameter zero and scale parameter ζ, the aggregate market share of labor choosing training path j is mapped by the standard multinomial conditional logit formulation (McFadden (1973)):

$$L_j^* = \frac{\exp\left((\beta\Omega_j - \bar{C}_j^*)/\zeta\right)}{\sum_{h=0}^{J} \exp\left((\beta\Omega_h - \bar{C}_h^*)/\zeta\right)} \tag{18}$$

2.3 Equilibrium

The equilibrium of the model has two layers. Within each period, the distribution of workers across training paths is predetermined, workers supply their labor inelastically, and the prices of the occupational goods adjust to clear the goods markets. Across generations, the young choose their training path anticipating the prices that will prevail when they work; since these prices depend on how many agents choose each path, the training choice must be consistent with the prices it generates.

Following Althoff and Reichardt (2026), occupational goods are combined by a CES aggregator with elasticity of substitution ϵ and importance weights α_j, so that the demand for good j is $D_j(\vec{p}) = \alpha_j p_j^{-\epsilon} E$, where E is aggregate expenditure. The CES price index $P = \left(\sum_{j=0}^{J} \alpha_j p_j^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}$ is the numéraire ($P = 1$), so that the expenditure share of good j is $s_j = \alpha_j p_j^{1-\epsilon}$. Aggregate expenditure equals total income, $E = \sum_{j=0}^{J} p_j L_j Y_j$, which is the sum of wages and payments to capital. Since preferences are homothetic, the distribution of income between workers and owners of capital does not affect relative demand. To close the capital side, I follow the quantitative implementation of Althoff and Reichardt (2026) and assume that the unit cost of producing an automatable task with capital relative to producing it with labor,

$$\chi \equiv \frac{R/\phi_\tau}{w_j/\gamma_\tau},$$

is constant across tasks, training paths and equilibria. Holding χ fixed means that the cost of AI in a task moves proportionally with the cost of the labor it replaces, i.e., that the effective productivity of capital ϕ_τ adjusts with wages. Notice that $\chi < 1$ is exactly the cost advantage of capital assumed in Section 2.1, since $\frac{R}{\phi_\tau} < \frac{w_j}{\gamma_\tau}$ if and only if $\chi < 1$; the calibrated value $\chi = 0.73$ (Section 4) therefore guarantees that all automatable tasks are produced with capital. As in Althoff and Reichardt (2026), I further assume that automated tasks are, on average, as productive as the tasks that remain performed by humans. Tasks that are automatable but not automated satisfy

this exactly, since by Assumption 1 they have $\gamma_\tau = 1$ like the automated ones; the assumption is therefore only an approximation for the augmented non-automatable tasks. Under these assumptions, the cost of capital relative to the output price, $\frac{R/\phi_\tau}{p_j}$, depends only on A_j, and the labor share is

$$\Gamma_j = \frac{\chi^{\rho-1}(1-A_j)}{A_j + \chi^{\rho-1}(1-A_j)},$$

which depends only on the set of automated tasks. Hence, output per worker Y_j does not depend on prices or on L_j.

Definition (Within-Period Equilibrium): Given the distribution of workers $\vec{L} = \{L_j\}_{j=0}^J$ and the realized AI shocks $\{A_j, B_j\}_{j=0}^J$, a within-period equilibrium is a set of allocations $\{l_{j,\tau}, k_{j,\tau}, Y_j\}$ and prices $\vec{p} = \{p_j\}_{j=0}^J$ such that:

1. Firms maximize profits: $l_{j,\tau}$, $k_{j,\tau}$ and Y_j are given by Equations (3), (4) and (5), and wages are given by Equation (6).

2. Goods markets clear: $L_j Y_j = \alpha_j p_j^{-\epsilon} E$ for every j, with $E = \sum_{h=0}^J p_h L_h Y_h$.

I denote by $p_j(\vec{L}, \vec{A}, \vec{B})$ the resulting equilibrium price. Since Y_j is independent of L_j, the goods market clearing condition implies that p_j is decreasing in L_j: more workers in path j raise the supply of good j and lower its price.

Definition (Post-AI Equilibrium): Given the average training costs $\{\bar{C}_j^*\}_{j=0}^J$ and the beliefs about AI shocks $\{\mu_{A,j}, \sigma_{A,j}^2, \mu_{B,j}, \sigma_{B,j}^2\}_{j=0}^J$, a post-AI equilibrium is a distribution of workers $\vec{L}^*$ such that:

1. Each young agent chooses the training path that solves Equation (17), so that L_j^* is given by Equation (18).

2. Expectations are rational: the certainty equivalent Ω_j in Equation (16) is evaluated at the price $p_j(\vec{L}^*, \vec{\mu}_A, \vec{\mu}_B)$ generated by the within-period equilibrium under the distribution of workers $\vec{L}^*$.

The post-AI equilibrium is therefore a fixed point $\vec{L}^* = \mathcal{L}\left(\vec{\Omega}(\vec{L}^*)\right)$, where $\mathcal{L}(\cdot)$ denotes the logit mapping in Equation (18). When the AI shocks are realized, the realized wages of the post-AI generation are given by the within-period equilibrium evaluated at $\vec{L}^*$ and at the realized $\{A_j, B_j\}$. The pre-AI steady state is the special case in which agents do not anticipate AI ($A_j = 0$ and $B_j = S_j^{non}$ with certainty, so that $\Pi_j = 1$), the risk premium is zero, and Ω_j equals the log-wage of the current working generation.

2.4 Theoretical Results and Equilibrium Properties

This section characterizes how the post-AI equilibrium of Section 2.3 responds to the uncertainty about the AI shocks. To simplify the notation, let

$$\Delta_{A,j} \equiv \left.\frac{\partial \ln w_j}{\partial A_j}\right|_{\mu_{A,j},\,\mu_{B,j}}, \qquad \Delta_{B,j} \equiv \left.\frac{\partial \ln w_j}{\partial B_j}\right|_{\mu_{A,j},\,\mu_{B,j}}$$

denote the sensitivities in Equation (11), so that Equation (12) reads $Var(\ln w_j) = \Delta_{A,j}^2\sigma_{A,j}^2 + \Delta_{B,j}^2\sigma_{B,j}^2$. Two features of the model make the comparative statics tractable. First, the sensitivities depend on the expected shocks and on the calibrated expenditure shares entering λ_j, but not on the variances or on the distribution of workers. Second, the expected price $p_j(\vec{L}, \vec{\mu}_A, \vec{\mu}_B)$ depends on the expected shocks but not on their variances. Hence, the variances $\sigma_{A,j}^2$ and $\sigma_{B,j}^2$ affect the equilibrium only through the risk premium in Equation (16). The expected shocks $\mu_{A,j}$ and $\mu_{B,j}$, instead, shift both the expected log-wage and the sensitivities: their effect on employment is theoretically ambiguous, because of the displacement, productivity and price effects discussed in Section 2.2, and I study it quantitatively in Section 5.

Proposition 1 (Wage Response to Labor Supply)**.** *In the within-period equilibrium, the price of good j responds to the distribution of workers according to*

$$\frac{\partial \ln p_j}{\partial \ln L_k} = \frac{s_k - \mathbf{1}[k=j]}{\epsilon}, \tag{19}$$

where s_k is the expenditure share of good k. Since output per worker Y_j does not depend on L_j, the expected log-wage is strictly decreasing in the number of workers in path j:

$$\frac{\partial\, \mathbb{E}[\ln w_j]}{\partial \ln L_j} = \frac{\partial \ln p_j}{\partial \ln L_j} = -\frac{1-s_j}{\epsilon} < 0.$$

Proof: Goods market clearing and the numéraire $P = 1$ imply $\ln p_j = \frac{1}{\epsilon}\left[\ln \alpha_j + \ln E - \ln(L_j Y_j)\right]$, where E equals aggregate real output, so that $d\ln E = \sum_k s_k\, d\ln(L_k Y_k)$. Since Y_k does not depend on the distribution of workers, Equation (19) follows. Finally, $\mathbb{E}[\ln w_j]$ depends on $\vec{L}$ only through $\ln p_j$. ■

Intuition: When more workers crowd into a training path, they collectively produce more of the associated good. The resulting excess supply depresses its price, and since wages equal the value of marginal product, expected wages fall. The reverse holds when workers leave a path: scarcity drives up the price and lifts expected wages. This mechanism is the central general-equilibrium channel through which the beliefs about the AI shocks translate into wage adjustments.

Proposition 2 (Flight from Uncertainty)**.** *In the post-AI equilibrium, an increase in automation uncertainty for training path j ($\uparrow \sigma^2_{A,j}$) decreases employment in path j and increases employment in all other paths $h \neq j$:*

$$\frac{\partial L^*_j}{\partial \sigma^2_{A,j}} = -\frac{\beta(\eta-1)}{2} \frac{\Delta^2_{A,j} L^*_j(1-L^*_j)}{\zeta + \beta/\epsilon} < 0, \tag{20}$$

$$\frac{\partial L^*_h}{\partial \sigma^2_{A,j}} = \frac{\beta(\eta-1)}{2} \frac{\Delta^2_{A,j} L^*_h L^*_j}{\zeta + \beta/\epsilon} > 0. \tag{21}$$

The same expressions hold for augmentation uncertainty ($\uparrow \sigma^2_{B,j}$), with $\Delta_{B,j}$ in place of $\Delta_{A,j}$.

Proof: Let $U_k \equiv \beta\Omega_k - \bar{C}^*_k$. Differentiating the logit shares in Equation (18) gives $\zeta\, d\ln L^*_k = dU_k - \sum_h L^*_h dU_h$. By Equation (16) and Proposition 1,

$$dU_k = \frac{\beta}{\epsilon}\left(\sum_m s_m\, d\ln L^*_m - d\ln L^*_k\right) - \mathbf{1}[k=j]\,\frac{\beta(\eta-1)}{2}\Delta^2_{A,j}\, d\sigma^2_{A,j}.$$

The term $\sum_m s_m\, d\ln L^*_m$ is common to all paths and cancels out, and $\sum_h L^*_h d\ln L^*_h = \sum_h dL^*_h = 0$. Therefore,

$$\left(\zeta + \frac{\beta}{\epsilon}\right) d\ln L^*_k = -\frac{\beta(\eta-1)}{2}\Delta^2_{A,j}\left(\mathbf{1}[k=j] - L^*_j\right) d\sigma^2_{A,j},$$

which yields Equations (20) and (21). The proof for $\sigma^2_{B,j}$ is identical. ■

Intuition: As a training path becomes more unpredictable due to exposure to AI, risk-averse individuals flee to safer alternatives. The magnitude of the flight increases with risk aversion (η) and with the sensitivity of the wage to the shock ($\Delta^2_{A,j}$ or $\Delta^2_{B,j}$), and it is largest for paths of intermediate size, as captured by $L^*_j(1-L^*_j)$.

Proposition 2 also quantifies the role of general equilibrium. Holding prices fixed, the response of employment would have ζ alone in the denominator. The term β/ϵ captures the price adjustment of Proposition 1: as workers leave path j, its price rises and makes it more attractive, which dampens the reallocation by a factor $\zeta/(\zeta+\beta/\epsilon)$. This factor lies in $(0,1)$, so general equilibrium effects dampen, but never reverse, the flight from uncertainty.

Proposition 3 (Endogenous Risk Premium)**.** *In the post-AI equilibrium, an increase in automation uncertainty for training path j ($\uparrow \sigma^2_{A,j}$) raises the expected log-wage in*

path j and lowers it in all other paths $h \neq j$:

$$\frac{\partial\, \mathbb{E}[\ln w_j]}{\partial \sigma^2_{A,j}} = \frac{1-s_j}{\epsilon}\,\frac{\beta(\eta-1)}{2}\,\frac{\Delta^2_{A,j}}{\zeta+\beta/\epsilon} > 0, \qquad \frac{\partial\, \mathbb{E}[\ln w_h]}{\partial \sigma^2_{A,j}} = -\frac{s_j}{\epsilon}\,\frac{\beta(\eta-1)}{2}\,\frac{\Delta^2_{A,j}}{\zeta+\beta/\epsilon} < 0. \tag{22}$$

The same expressions hold for $\sigma^2_{B,j}$, with $\Delta_{B,j}$ in place of $\Delta_{A,j}$.

Proof: Since the variances do not affect the expected price directly, $d\,\mathbb{E}[\ln w_k] = d\ln p_k = \frac{1}{\epsilon}\left(\sum_m s_m\, d\ln L^*_m - d\ln L^*_k\right)$ by Proposition 1. From the proof of Proposition 2, $\sum_m s_m\, d\ln L^*_m = -\frac{\beta(\eta-1)}{2}\frac{\Delta^2_{A,j}}{\zeta+\beta/\epsilon}(s_j - L^*_j)\, d\sigma^2_{A,j}$. Substituting this expression and $d\ln L^*_k$ yields Equation (22). ■

Intuition: Uncertainty reduces the share of workers choosing training path j (Proposition 2). The resulting labor scarcity pushes up the equilibrium price of the associated good, increasing the expected wage of those who remain, while the workers who move to other paths push down their wages. This is a compensating differential for risk (Rosen, 1986), generated endogenously by the reallocation of the new generation across training paths: there is no such premium in the pre-AI equilibrium, and it emerges only because AI makes some training paths riskier than others. The compensation is only partial: since $\frac{\beta(1-s_j)}{\epsilon} < \zeta + \frac{\beta}{\epsilon}$, the increase in $\mathbb{E}[\ln w_j]$ is smaller than the increase in the risk premium $\frac{\eta-1}{2}\Delta^2_{A,j}\sigma^2_{A,j}$, so that the certainty equivalent Ω_j still falls.

3 Measuring the AI Shocks

3.1 Measuring the AI Shocks: Methodology

To implement the model quantitatively, we need to measure, for each training path j, the moments of the two aggregate shocks introduced in Section 2.2: the mean and variance of the automation share A_j ($\mu_{A,j}$ and $\sigma^2_{A,j}$) and of the augmentation index B_j ($\mu_{B,j}$ and $\sigma^2_{B,j}$), together with the shares $S^{cur}_{A,j}$, $S^{pot}_{A,j}$ and S^{non}_j that enter the expected augmentation factor $\mu_{\Pi,j}$. I compute these objects by classifying the tasks of each training path according to their current and potential AI exposure.

This classification relies on two main sources: the actual task penetration of AI (Massenkoff and McCrory, 2026), and the theoretical automatability and augmentation potential of tasks (Althoff and Reichardt, 2026).

Massenkoff and McCrory (2026) construct a measure of current task penetration of AI using recent Anthropic data. The task penetration is zero if, at the moment, AI is not used in that task. If instead AI is already used in that task, its penetration is positive, with values closer to 1 meaning that AI is used more to automate than to augment that task (see Massenkoff and McCrory (2026) for more details).

Althoff and Reichardt (2026) provide two measures: theoretical automatability and theoretical augmentation. Theoretical automatability equals 1 if the task is potentially automatable using generative AI, and 0 otherwise. Theoretical augmentation is the potential augmentation factor g_τ of Assumption 1: it equals 1 if the task is not augmentable with AI and is greater than 1 if it is potentially augmentable; for example, $g_\tau = 1.1$ means that AI can boost labor productivity by 10% in that task (see Althoff and Reichardt (2026) for more details).

First, I define the *current status* of a task:

- **Automated:** The actual task penetration is greater than or equal to 0.9 and the task is automatable (automatable = 1).
- **Augmented:** The actual task penetration is greater than 0 and the task is not automatable (automatable = 0). Alternatively, the task is automatable (automatable = 1), but the current task penetration is positive and less than 0.9.
- **Neither:** If the actual task penetration is zero, the task is currently neither automated nor augmented.

Second, I map the current status and the theoretical measures into the three sets of Section 2.2, which determine the *potential status* of each task:

- **Already automated** ($\mathcal{S}_{A,j}^{cur}$): tasks that are currently automated. Their future status is automated with certainty.
- **Potentially automatable** ($\mathcal{S}_{A,j}^{pot}$): tasks that are theoretically automatable (automatable = 1) but not yet automated (actual penetration below 0.9). By Assumption 1(i), these tasks face only automation risk: if they are not automated, their labor productivity is $\gamma_\tau = 1$, even when they are currently augmented.
- **Not automatable** ($\mathcal{S}_{j}^{non}$): tasks that are not theoretically automatable (automatable = 0). These tasks are never automated, but they can be augmented. Within this set, I further distinguish:
  - **Already augmented**: tasks with positive actual penetration. Their future status is augmented with certainty, $\gamma_\tau = g_\tau$.
  - **Potentially augmentable**: tasks with zero actual penetration and $g_\tau > 1$. Their augmentation is uncertain.
  - **Not affected by AI**: tasks with zero actual penetration and $g_\tau = 1$, so that $\gamma_\tau = 1$.

Let $\mathcal{T}_j$ be the set of tasks associated with training path j, with importance weights $\theta_{j,\tau}$ such that $\sum_{\tau\in\mathcal{T}_j}\theta_{j,\tau}=1$, and recall that $a_\tau\in\{0,1\}$ and $b_\tau\in\{0,1\}$ indicate whether task τ will eventually be automated and augmented, respectively. The shares in Section 2.2 are $S^{cur}_{A,j}=\sum_{\tau\in\mathcal{S}^{cur}_{A,j}}\theta_{j,\tau}$, $S^{pot}_{A,j}=\sum_{\tau\in\mathcal{S}^{pot}_{A,j}}\theta_{j,\tau}$ and $S^{non}_{j}=\sum_{\tau\in\mathcal{S}^{non}_{j}}\theta_{j,\tau}$.

Automation. If a task is already automated, its future outcome is certain: $\mathbb{E}[a_\tau]=1$ and $Var(a_\tau)=0$. Conversely, if it is not automatable, $\mathbb{E}[a_\tau]=0$ and $Var(a_\tau)=0$. The uncertainty lies entirely with the tasks that are *potentially automatable* but not yet automated. For these tasks in the "gap", I assume a probability $p_A\in(0,1)$ that the theoretical automatability translates into actual automation, so that $\mathbb{E}[a_\tau]=p_A$ and $Var(a_\tau)=p_A(1-p_A)$. The expected automation share for training path j is the weighted sum of these task-level expectations:

$$\mu_{A,j}=\sum_{\tau\in\mathcal{T}_j}\theta_{j,\tau}\mathbb{E}[a_\tau]=S^{cur}_{A,j}+p_A S^{pot}_{A,j}. \tag{23}$$

To allow AI automation to be correlated across tasks within the same training path, I introduce a correlation parameter $\kappa\in[0,1]$. A value of $\kappa=0$ implies independent task automation, whereas $\kappa=1$ implies perfect correlation (AI automates either all potential tasks or none). The variance of the automation share is then a convex combination of the independent and perfectly correlated variances:

$$\sigma^2_{A,j}=p_A(1-p_A)\left[(1-\kappa)\sum_{\tau\in\mathcal{S}^{pot}_{A,j}}\theta^2_{j,\tau}+\kappa\left(S^{pot}_{A,j}\right)^2\right]. \tag{24}$$

Two features are worth noting. First, the variance is strictly increasing in the automation gap $S^{pot}_{A,j}$; therefore, training paths with a higher unrealized potential for automation face higher uncertainty. Second, the term $\sum\theta^2_{j,\tau}$ acts as a Herfindahl-Hirschman Index (HHI) of task concentration. This implies that, holding the gap share constant, training paths that are highly specialized in a few uncertain tasks face significantly more risk than paths with a diversified portfolio of many small tasks.

Augmentation. By Assumption 1, augmentation only concerns the non-automatable tasks, and it enters the wage through the augmentation index $B_j=\sum_{\tau\in\mathcal{S}^{non}_{j}}\theta_{j,\tau}\gamma_\tau^{\rho-1}$, where $\gamma_\tau=1+b_\tau(g_\tau-1)$. Since b_τ is Bernoulli, $\gamma_\tau^{\rho-1}=1-b_\tau\left(1-g_\tau^{\rho-1}\right)$, and it is convenient to define the task-level augmentation gain $d_\tau\equiv 1-g_\tau^{\rho-1}\in[0,1)$, which is zero for tasks that are not augmentable and increases with g_τ. For tasks that are already augmented, $\mathbb{E}[b_\tau]=1$ and $Var(b_\tau)=0$; for tasks that are not affected by AI, $d_\tau=0$. The uncertainty lies with the tasks that are *potentially augmentable*, which I denote by $\mathcal{S}^{pot}_{B,j}\subseteq\mathcal{S}^{non}_{j}$: for these tasks, I assume a probability $p_\gamma\in(0,1)$ that the

potential augmentation is realized, so that $\mathbb{E}[b_\tau] = p_\gamma$ and $Var(b_\tau) = p_\gamma(1 - p_\gamma)$. The expected augmentation index is

$$\mu_{B,j} = \sum_{\tau \in \mathcal{S}_j^{non}} \theta_{j,\tau} \left(1 - \mathbb{E}[b_\tau]\, d_\tau\right) = S_j^{non} - \sum_{\tau \in \mathcal{S}_{B,j}^{cur}} \theta_{j,\tau} d_\tau - p_\gamma \sum_{\tau \in \mathcal{S}_{B,j}^{pot}} \theta_{j,\tau} d_\tau, \tag{25}$$

where $\mathcal{S}_{B,j}^{cur}$ denotes the set of already augmented tasks. Recall that a lower B_j corresponds to more augmentation: $\mu_{B,j} = S_j^{non}$ when no task is augmented, and the expected augmentation factor that enters the wage is $\mu_{\Pi,j} = 1 - S_j^{non} - \mu_{A,j} + \mu_{B,j}$. Allowing for the same within-path correlation κ, the variance of the augmentation index is

$$\sigma_{B,j}^2 = p_\gamma(1 - p_\gamma) \left[(1 - \kappa) \sum_{\tau \in \mathcal{S}_{B,j}^{pot}} \theta_{j,\tau}^2 d_\tau^2 + \kappa \left(\sum_{\tau \in \mathcal{S}_{B,j}^{pot}} \theta_{j,\tau} d_\tau \right)^2 \right]. \tag{26}$$

Just as with automation, this variance is driven by two key factors: the size of the potential augmentation gap, weighted by the potential productivity gains d_τ, and the concentration of that potential across a few specific tasks. Consistent with Assumption 1(ii), the automation and augmentation shocks are drawn independently, so that the two variances can be computed separately.

3.2 Measuring the AI Shocks: Results

For the baseline empirical implementation, I set the probability of automation $p_A = 0.5$ and the probability of augmentation $p_\gamma = 0.5$. This means that if automation or augmentation is theoretically possible but not yet realized, agents believe there is a 50% chance that it will actually occur. While this is a simplification, it captures the idea that a young student is agnostic about the future realization of AI tasks. The correlation parameter is set to $\kappa = 0.5$, the midpoint between the two unrealistic extremes of independence ($\kappa = 0$) and perfect correlation ($\kappa = 1$). I also consider an alternative specification with higher probabilities, $p_A = 0.8$ and $p_\gamma = 0.8$, whose implications for employment and wages are discussed in Section 6.

Figure 2 displays the moments of the two shocks in the benchmark. The left panel reports the expected automation share $\mu_{A,j}$. The right panel reports the expected augmentation gain $S_j^{non} - \mu_{B,j}$, which is zero if no task is augmented and increases with the number and the potential productivity gains d_τ of the augmented tasks. Since $\mu_{\Pi,j} = 1 - \mu_{A,j} - (S_j^{non} - \mu_{B,j})$, the two panels are measured in the same units: an increase of 0.01 in either of them lowers the expected augmentation factor by 0.01. The two shocks differ, however, in their effect on wages: automation also lowers the labor share Γ_j (displacement effect), while augmentation only raises output per worker.

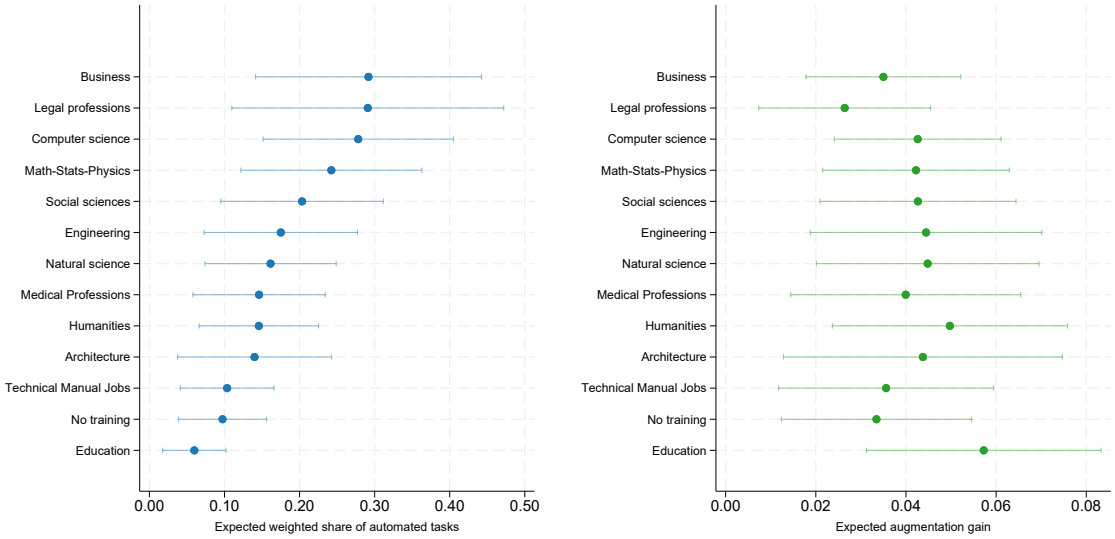


Figure 2: Expected Automation Share and Augmentation Gain across Training Paths (± 1 standard deviation).

Notes: For each training path j, the left panel reports the expected automation share $\mu_{A,j}$ (Equation (23)), and the right panel the expected augmentation gain $S_j^{non} - \mu_{B,j}$, i.e., the expected reduction of the augmentation index B_j below its no-augmentation value S_j^{non} (Equation (25)). Both objects are measured in units of task weight and lower the expected augmentation factor one-for-one, $\mu_{\Pi,j} = 1 - \mu_{A,j} - (S_j^{non} - \mu_{B,j})$. Dots are expected values and error bars span ± 1 standard deviation ($\sigma_{A,j}$ and $\sigma_{B,j}$, Equations (24) and (26)).

Expected automation varies considerably across training paths. The most exposed paths are legal professions and business, where almost 30% of the weighted tasks are expected to be automated, followed by computer science (28%) and math, statistics and physics (24%). At the other extreme, expected automation is 6% in education and around 10% for technical manual jobs and no training. Uncertainty follows a similar ranking, but it is driven by the automation gap $S^{pot}_{A,j}$ rather than by the expected share itself. Legal professions face the largest uncertainty ($\sigma_{A,j} = 0.18$), because half of their weighted tasks are potentially automatable but not yet automated, followed by business (0.15). Computer science has almost the same expected automation as these two paths, but a lower uncertainty (0.13): about 10% of its weighted tasks are already automated (the highest share across training paths), and the realized part of its exposure is no longer uncertain. In general, the training paths most exposed to automation are those that are more abstract and cognitive (legal professions, business, computer science), while the least exposed are those that require manual and physical skills (technical manual jobs, no training) or social interactions (education).

Expected augmentation gains are smaller and much less dispersed across training paths, ranging from 0.026 in legal professions to 0.057 in education, with standard deviations between 0.017 and 0.031. Although small in absolute terms, these gains are economically relevant: at $\mu_{\Pi,j} \approx 0.8$, a gain of 0.04 raises output per worker by roughly $\frac{1}{1-\rho}\frac{0.04}{0.8} \approx 10\%$. Since, by Assumption 1(i), only the non-automatable tasks can be augmented, augmentation tends to be lower where automation is higher: legal professions and business have the lowest expected augmentation gains, while education has the highest, also because 18% of its weighted tasks are already augmented and contribute to the gain with certainty. Because both the expected gains and their uncertainty are similar across training paths, augmentation mostly shifts wages in all paths in the same direction, and its role in shaping the relative attractiveness of training paths is more limited than that of automation.

Lastly, Figure 3 compares the benchmark with the alternative scenario where $p_A = 0.8$ and $p_\gamma = 0.8$. Expected automation increases in all training paths by $0.3\, S^{pot}_{A,j}$, so that the increase is largest where the automation gap is largest (e.g., from 0.29 to 0.44 in legal professions, compared with an increase from 0.06 to 0.09 in education), and differences across paths widen. The expected augmentation gains also increase in all paths. Uncertainty, instead, is lower: the Bernoulli variance $p(1-p)$ falls from 0.25 to 0.16, so that all standard deviations shrink by 20%. Hence, in the alternative scenario, agents expect more AI but are less uncertain about it, and the ranking of training paths is almost unchanged.

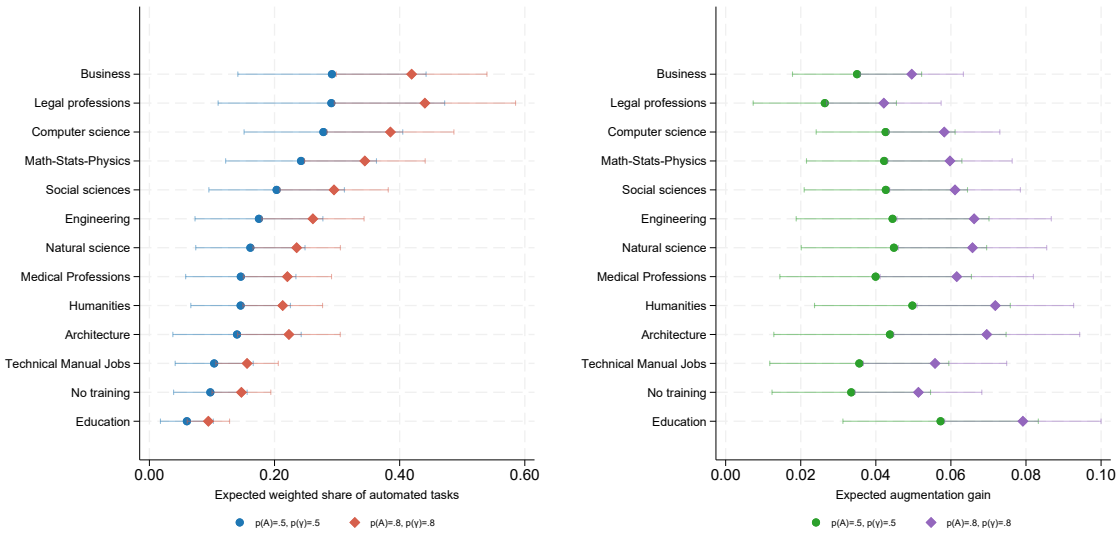


Figure 3: Expected Automation Share and Augmentation Gain across Training Paths (± 1 standard deviation): Benchmark vs. Alternative Scenario.

Notes: Same objects as in Figure 2: expected automation share $\mu_{A,j}$ (left panel) and expected augmentation gain $S_j^{non} - \mu_{B,j}$ (right panel), with error bars spanning ± 1 standard deviation ($\sigma_{A,j}$ and $\sigma_{B,j}$). Circles (blue and green) refer to the benchmark, where potentially automatable and potentially augmentable tasks are realized with probability $p_A = p_\gamma = 0.5$; diamonds (orange and purple) refer to the alternative scenario with $p_A = p_\gamma = 0.8$.

4 Calibration

The quantitative implementation of the model proceeds in two main stages. First, I calibrate the model to a pre-AI baseline equilibrium (representing the year 2022) where AI shocks are not yet anticipated. This allows me to recover the structural parameters of the economy that are not directly observed: the average training costs $\bar{C}_j^*$ and the demand weights α_j. Second, I introduce the moments of the AI shocks measured in Section 3 ($\mu_{A,j}$, $\sigma^2_{A,j}$, $\mu_{B,j}$ and $\sigma^2_{B,j}$) to simulate the post-AI equilibrium of Section 2.3, and I decompose its effects on employment and wages. The main assumption is that AI does not affect the structural parameters of the economy (training costs, preferences and elasticities). This is a strong assumption, but as long as AI does not drastically change these parameters, at least in relative terms, the implications of the model are robust to it.

4.1 Training Paths, Employment and Wages

I consider 13 training paths: eleven cognitive paths, corresponding to fields of college degree (natural science, architecture, humanities, computer science, education, engineering, legal professions, medical professions, math-statistics-physics, social sciences

and business), one technical path (technical manual jobs) and the no-training option ($j = 0$). Each cognitive path groups the 2-digit fields of the Classification of Instructional Programs (CIP), and it is linked to the occupations (6-digit SOC codes) of the official NCES CIP-SOC crosswalk. The technical path and the no-training option collect the occupations that the crosswalk links to vocational programs (e.g., culinary and personal services, protective services) or to no specific program, and they are separated using the typical on-the-job training required by each occupation according to the BLS Employment Projections: occupations that require moderate-term or long-term on-the-job training or an apprenticeship are assigned to technical manual jobs, while those that require short-term or no on-the-job training are assigned to no training.

Within each training path, occupation k receives a weight $\omega_{j,k}$ equal to its share of the employment of the path. For the cognitive paths, I compute these weights from the 2022 IPUMS American Community Survey (ACS), counting the employed workers older than 30 (who have completed their training) with a degree in the fields of path j who work in occupation k; only the degree-occupation pairs included in the official crosswalk are kept. For the technical path and the no-training option, I use the 2022 BLS employment of each occupation. The same weights enter the task weights of Section 3, $\theta_{j,\tau} = \sum_k \omega_{j,k}\,\theta_{k,\tau}$, where $\theta_{k,\tau}$ is the importance of task τ in occupation k from Althoff and Reichardt (2026), normalized to sum to one within each occupation.

The pre-AI wage of path j, w_j^{2022}, is the $\omega_{j,k}$-weighted average of the 2022 BLS median hourly wages of its occupations. The pre-AI employment shares L_j^{2022} are built in two steps. First, the share of workers in the cognitive paths is the weighted ACS share of employed workers older than 30 who hold a bachelor's degree (or higher) in one of the fields of these paths, which is 40.6%.[1] This share is split across the cognitive paths according to the ACS distribution of these workers across fields of degree. Second, the remaining 59.4% is split between technical manual jobs and no training according to the BLS employment of the occupations assigned to each of them. Finally, the expenditure shares s_j that enter the pass-through λ_j in Equation (11) are the 2022 wage shares, $s_j = w_j^{2022} L_j^{2022} / \sum_h w_h^{2022} L_h^{2022}$. Since the labor share is one in the pre-AI equilibrium (see below), the wage bill of path j coincides with the revenue of good j, and wage shares are exactly expenditure shares.

[1] In the ACS, the field of degree is reported only for workers with at least a bachelor's degree, and item nonresponse is imputed by the Census Bureau. Workers with a degree in a vocational field (cosmetology and culinary arts, criminal justice and fire protection) are not assigned to the cognitive paths. Including them, the share of employed workers older than 30 with a bachelor's degree is 41.4%.

4.2 Pre-AI Baseline Equilibrium (2022)

In the pre-AI steady state, agents do not anticipate the arrival of AI. Consequently, they expect no automation and no augmentation ($A_j = 0$ and $B_j = S_j^{non}$ with certainty, so that $\Pi_j = 1$), and there is no uncertainty regarding these shocks ($\sigma^2_{A,j} = \sigma^2_{B,j} = 0$). The labor share is therefore $\Gamma_j = 1$ and output per worker is $Y_j = 1$ (Equation (5)), so that the wage equation (6) reduces to $w_j = p_j$. Without uncertainty, the risk premium is zero, and the certainty equivalent collapses to the log-wage: $\Omega_j^{pre} = \ln w_j^{2022}$.

Using Equation (18), the employment share of training path j in the pre-AI equilibrium is

$$L_j^{2022} = \frac{\exp((\beta \ln w_j^{2022} - \bar{C}_j^*)/\zeta)}{\sum_{h=0}^{J} \exp((\beta \ln w_h^{2022} - \bar{C}_h^*)/\zeta)} \tag{27}$$

Given the observed employment shares and wages, I can invert this mapping to exactly recover the average training costs $\bar{C}_j^*$. Since the logit shares depend only on differences in costs, $\bar{C}_j^*$ is identified up to a common constant, and I normalize the cost of the no-training option to zero ($\bar{C}_0^* = 0$). The training costs are then

$$\bar{C}_j^* = \beta(\ln w_j^{2022} - \ln w_0^{2022}) - \zeta(\ln L_j^{2022} - \ln L_0^{2022}) \tag{28}$$

A training path has a high cost if its wage premium over no training is high relative to the share of workers who choose it.

On the demand side, I recover the importance weights α_j of the CES aggregator of Section 2.3. Since $s_j = \alpha_j p_j^{1-\epsilon}$ and $p_j = w_j^{2022}$ in the pre-AI equilibrium, for any two training paths i and j

$$\frac{\alpha_i}{\alpha_j} = \left(\frac{w_i^{2022}}{w_j^{2022}}\right)^{\epsilon-1} \frac{s_i}{s_j} = \left(\frac{w_i^{2022}}{w_j^{2022}}\right)^{\epsilon} \frac{L_i^{2022}}{L_j^{2022}} \tag{29}$$

and the level of the weights is pinned down by the numéraire, $P = \left(\sum_j \alpha_j (w_j^{2022})^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}} = 1$. By construction, the calibrated model reproduces the observed 2022 employment shares and wages exactly.

4.3 Externally Calibrated Parameters

The remaining parameters are taken from the literature on task-based models. Table 1 summarizes the benchmark values.

Elasticities. I set the elasticity of substitution across tasks within a training path to $\rho = 0.49$, as estimated by Humlum (2021), implying that tasks are complements, and the elasticity of substitution across the goods of different training paths to $\epsilon = 1.57$, the midpoint between 1.81 (Burstein et al., 2019) and 1.34 (Caunedo et al., 2023), as

in Althoff and Reichardt (2026). With this value, $\epsilon > 1 - s_j$ for all training paths, so that the pass-through λ_j lies in $(0, 1)$ (Section 2.2).

Preferences. The discount factor between the training and the working period is $\beta = 1.03^{-7} = 0.813$, which corresponds to an annual discount rate of 3% over a training period of 7 years (from age 18 to 25). The coefficient of relative risk aversion is set to $\eta = 5$ in the benchmark; in Section 6, I test a low ($\eta = 2$) and a high ($\eta = 10$) risk aversion scenario.

Responsiveness of training choices. The Gumbel dispersion ζ captures the rigidity of the training choice: a lower ζ means that agents are more responsive to differences in certainty equivalents and training costs, while a higher ζ means that idiosyncratic preferences matter more. Since there is no direct estimate of ζ (as defined in the model of this paper) for the choice of a training path, I discipline it through the elasticity of training choices to wages. From Equation (18), holding wages fixed, the elasticity of the share of path j with respect to its certainty-equivalent wage is

$$e_j \equiv \frac{\partial \ln L_j^*}{\partial \Omega_j} = \frac{\beta \left(1 - L_j^*\right)}{\zeta} \approx \frac{\beta}{\zeta},$$

so that a target elasticity e maps into $\zeta \approx \beta/e$. The relevant object is the partial equilibrium elasticity, since empirical estimates measure how students respond to the wages they observe, taking them as given. The literature offers values of e that differ by more than an order of magnitude, and none of them refers exactly to the choice studied in this paper. I therefore consider three scenarios, each disciplined by a different strand of evidence, and treat them as equally plausible.

High responsiveness ($\zeta = 0.053$). The first scenario uses the value estimated by Althoff and Reichardt (2026) by maximum likelihood on NLSY79 data, which implies an elasticity of about $0.813/0.053 \approx 15$: training choices respond so strongly that differences in certainty equivalents net of training costs are almost arbitraged away. Its strength is its consistency with the task-based framework from which the AI shocks are taken, and it captures the possibility that a large, permanent, and widely publicized technological shift triggers a much stronger response than the one observed in normal times. Its limitation is that the parameter has a different interpretation in their model: it is the scale of yearly productivity shocks in the repeated choice of occupation of workers who differ in their multidimensional skills, whereas in my model agents do not differ in their skills and the Gumbel shock absorbs all the heterogeneity in training choices.

Medium responsiveness ($\zeta = 0.5$). The second scenario uses the evidence from Roy models in which workers draw Fréchet-distributed productivities across occupations or sectors. In these models, the Fréchet shape parameter is the elasticity of the share

of workers in an occupation or sector with respect to its wage per efficiency unit, the counterpart of e_j above. Burstein et al. (2019) estimate it at 1.81 from the reallocation of US workers across occupations and equipment types between 1984 and 2003, and at 1.26 when controlling for group-specific time trends. Galle et al. (2023) obtain values between 1.42 and 2.79 from the response of sectoral employment to the China shock across US commuting zones, with a preferred value of 1.5. Hsieh et al. (2019) set it at 2, a compromise between the values implied by the dispersion of wages within occupations (1.52) and by the extensive-margin elasticity of labor supply (2.57). Since the preferred estimates lie between about 1.3 and 2, I take an elasticity of about 1.6, which gives $\zeta = 0.813/1.6 \approx 0.5$. Its strength is that the idiosyncratic term is a field-specific productivity, which plausibly accounts for part of the heterogeneity in training choices, and that the estimates describe the reallocation of the whole workforce over horizons of five to ten years, rather than year-to-year fluctuations. Its limitation is that these estimates refer to the allocation of workers across occupations or sectors and abstract from the persistent preferences for fields of study.

Low responsiveness ($\zeta = 1.2$). The third scenario targets the evidence on the choice of college majors. Long et al. (2015) estimate an elasticity of completed college majors in the US to lagged major-specific wage premia of 0.67, which gives $\zeta = 0.813/0.67 \approx 1.2$. Its strength is that it refers to the choice of a field of study, and it is in line with the broader finding that non-pecuniary factors are a key determinant of major choice (Beffy et al., 2012; Wiswall and Zafar, 2015). Its limitation is that it measures the response to observed, year-to-year fluctuations in wage premia, which may understate the response to a large and persistent shift in the returns to training such as the one induced by AI.

Since there are no grounds for preferring one scenario over the others, I report the main results as ranges across the three scenarios. To keep the exposition transparent, I use the high-responsiveness scenario as the benchmark for the detailed decomposition of the AI shock in Section 5, because it is consistent with the framework of Althoff and Reichardt (2026) and makes each channel most visible; the net effects under the three scenarios are compared in Section 5.3. The choice of the benchmark does not reflect a view that it is more likely than the other two. When ζ changes, the training costs are recalibrated with Equation (28), so that the model still reproduces the 2022 equilibrium.

AI shocks. The moments of Section 3 depend on the probabilities that potentially automatable and potentially augmentable tasks are eventually automated or augmented. In the benchmark, I set $p_A = p_\gamma = 0.5$. This choice reflects an agnostic view about the future of these tasks, and it maximizes the task-level uncertainty, since the Bernoulli variance $p(1-p)$ is largest at $p = 0.5$. In Section 6, I also consider an alternative scenario with $p_A = p_\gamma = 0.8$. The within-path correlation of AI success is $\kappa = 0.5$,

a task is classified as already automated if its current AI penetration is at least 0.9, and the potential augmentation factors g_τ are taken from Althoff and Reichardt (2026).

Cost of automation. Following Althoff and Reichardt (2026), I assume that using AI instead of labor in an automatable task saves 27% of the cost, the estimate of Acemoglu (2025) based on experimental evidence, which yields a relative cost of capital $\chi = 0.73$. Since $\chi < 1$, all automated tasks are produced with capital (Section 2.3), and the labor share of training path j depends only on its automation share,

$$\Gamma_j = \frac{\chi^{\rho-1}(1 - A_j)}{A_j + \chi^{\rho-1}(1 - A_j)},$$

which equals one in the pre-AI equilibrium ($A_j = 0$) and is evaluated at $A_j = \mu_{A,j}$ in the expected log-wage of the post-AI equilibrium.

Table 1: Calibrated Parameters.

Parameter	Description	Value	Source
ρ	Elasticity of substitution across tasks	0.49	Humlum (2021)
ϵ	Elasticity of substitution across goods	1.57	Burstein et al. (2019); Caunedo et al. (2023)
χ	Cost of capital relative to labor in automated tasks	0.73	Acemoglu (2025)
ζ	Dispersion of idiosyncratic training preferences:		
	high responsiveness (benchmark)	0.053	Althoff and Reichardt (2026)
	medium responsiveness	0.5	Burstein et al. (2019); Hsieh et al. (2019); Galle et al. (2023)
	low responsiveness	1.2	Long et al. (2015)
β	Discount factor over the training period	0.813	3% annual rate, 7 years
η	Relative risk aversion	5	Sensitivity with: 2, 10
p_A, p_γ	Probability of automation / augmentation	0.5	Sensitivity with: 0.8
κ	Within-path correlation of AI success	0.5	Midpoint of $[0, 1]$
$\bar{C}_j^*$	Average training costs	Equation (28)	2022 employment and wages
α_j	CES demand weights	Section 4.2	2022 employment and wages

4.4 Simulating the Post-AI Equilibrium

Once the structural parameters are pinned down, I introduce the AI shocks. For each training path j, I plug the expected shocks $(\mu_{A,j}, \mu_{B,j})$, together with S_j^{non}, into the expected log-wage, and the variances $(\sigma^2_{A,j}, \sigma^2_{B,j})$ into the variance of the log-wage (Equation (12)), with the sensitivities of Equation (11) evaluated at the expected shocks. The post-AI equilibrium is the fixed point $\vec{L}^* = \mathcal{L}(\vec{\Omega}(\vec{L}^*))$ of Section 2.3: the shares L_j^* chosen by the young generation (Equation (18)) determine the supply of each good and hence the equilibrium prices $p_j(\vec{L}^*, \vec{\mu}_A, \vec{\mu}_B)$, which in turn determine the certainty equivalents Ω_j that the young generation takes into account. I solve for this fixed point numerically, keeping the training costs $\bar{C}_j^*$ and the demand weights α_j at their calibrated values.

By comparing the post-AI equilibrium with the pre-AI baseline, I quantify the total effect of AI on employment shares and expected wages. Employment effects are reported as percentage changes in the employment share L_j. Wage effects are reported as percentage changes in the expected real wage, $\exp(\Delta\mathbb{E}[\ln w_j]) - 1$, i.e., in the geometric mean of the wage of path j; since the CES price index is the numéraire, these are changes in real terms. To isolate the contribution of each channel, I introduce the four components of the AI shock sequentially, solving for a new equilibrium at each step:

1. *Expected automation:* $A_j = \mu_{A,j}$, with no augmentation ($B_j = S_j^{non}$) and no uncertainty.
2. *Expected augmentation:* in addition, $B_j = \mu_{B,j}$.
3. *Automation uncertainty:* in addition, $\sigma^2_{A,j}$ enters the risk premium.
4. *Augmentation uncertainty:* in addition, $\sigma^2_{B,j}$ enters the risk premium. This step yields the full post-AI equilibrium.

The contribution of each channel is the difference between two consecutive equilibria. For the wage effects of the expected shocks, I also report the change in the expected log-wage holding the distribution of workers fixed at $\vec{L}^{2022}$, which isolates the direct effect of AI from the general equilibrium effect of labor reallocation.

5 Results and Decomposition

This section presents the quantitative predictions of the calibrated model. To isolate the contribution of each economic mechanism, the full post-AI equilibrium is decomposed

sequentially into four channels: the *expected automation* effect, the *expected augmentation* effect, and the *uncertainty* effects associated with both automation and augmentation shocks. Sections 5.1 and 5.2 present the decomposition in the benchmark (high-responsiveness) scenario, where each channel is most visible. Section 5.3 then compares the net effects across the three equally plausible scenarios for the responsiveness of training choices, and shows which results depend on it.

5.1 Employment Reallocation: High-Responsiveness Scenario

Figure 4 presents the decomposition of AI-induced employment shifts across the 13 training paths in the high-responsiveness scenario.

The *expected automation* channel drives the primary structural transformation. The training paths most exposed to automation—such as business, computer science, and legal professions—are expected to be chosen less by the new generation of workers, with legal professions losing 12% of their workforce only because of the automation threat. Conversely, paths with high physical or interpersonal requirements—such as technical manual jobs and education, as well as the "no training" residual category—experience positive labor inflows. The *expected augmentation* channel plays a comparatively minor role in employment reallocation, as human-enhancing technologies boost output across most sectors relatively symmetrically without dramatically shifting relative labor demand.

Moreover, the *uncertainty* over automation triggers a secondary reallocation. Risk-averse agents actively flee the highly uncertain training paths (e.g., business and legal professions). Although this impact is relatively smaller, it slightly amplifies the initial expected automation shock, accelerating the employment drop in exposed training paths and crowding workers into the options perceived as safer. Uncertainty regarding augmentation has virtually no impact on employment, again because the augmentation uncertainty is relatively similar across training paths.

Overall, in the benchmark the model predicts a substantial reallocation of labor across training paths, with the most exposed paths (legal professions, business, and computer science) losing between 11% and 16% of their workforce, while the safer paths (technical manual jobs, education, and no training) gain up to +8%. As shown in Section 5.3, the size of this reallocation scales with the responsiveness of training choices, while its direction and the ranking of training paths do not.

5.2 Wage Effects: High-Responsiveness Scenario

The wage decomposition (Figure 5) reveals the power of general equilibrium forces. The expected AI shocks—both automation and augmentation—have a positive impact on

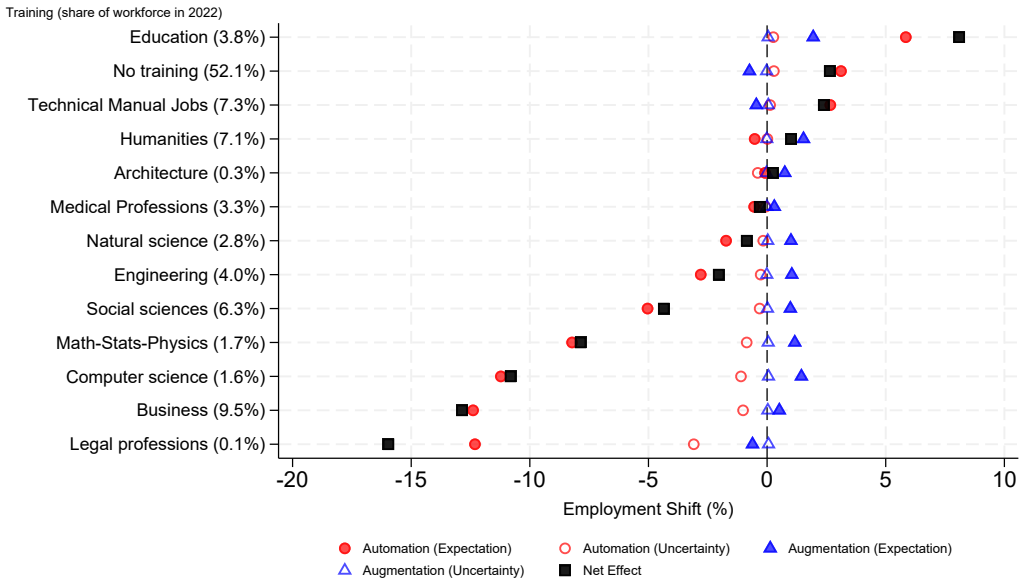


Figure 4: Decomposition of AI-induced employment shifts across training paths.

Notes: This figure decomposes, in the benchmark (high-responsiveness, $\zeta = 0.053$) scenario, the percentage change in employment shares (relative to the pre-AI 2022 baseline) into four channels: expected automation (red solid circles), uncertainty over automation (red hollow circles), expected augmentation (blue solid triangles), and uncertainty over augmentation (blue hollow triangles). The total net effect is shown as black squares. See Section 4 for details on the benchmark calibration. See Section 3 for details on the expected AI shocks and their uncertainty.

aggregate productivity, generating an across-the-board increase in expected wages ranging from +4.1% to +5.4% from automation, and of about +10% from augmentation. The primary driver of this result is the reallocation of labor across training paths. If we hold labor supply fixed at the pre-AI baseline, the expected wage impact of automation varies dramatically across paths. Training paths highly exposed to automation (legal, business, and computer science) would see their expected wages drop by 2% to 3%, whereas the safer paths (technical manual jobs, education, and no training) would gain up to +11% only because of expected automation. This is intuitive: automation boosts productivity in the non-automated tasks; therefore, if workers do not reallocate, training paths that are less exposed to automation would gain a lot (low displacement effect), whereas the more exposed paths would lose because of the displacement effect. The fact that individuals reallocate toward positively impacted paths, however, erodes the expected wage gains in those paths and generates a scarcity effect in the exposed paths. In the benchmark, where training choices are highly responsive, this reallocation makes the expected wage gains very similar across all training paths. These results highlight the importance of accounting for the labor supply responses when evaluating the impact of AI on wages.

The impact of uncertainty is qualitatively similar. Risk-averse agents flee the highly uncertain training paths, which reduces employment in those paths. The labor scarcity generated by this flight to safety, however, creates an endogenous risk premium that drives up expected wages in the exposed paths. Quantitatively, this channel is less pronounced, but it is still economically significant. In the benchmark, legal professions, the most uncertain training path, see their expected wages increase by +2.2% only because of the uncertainty-driven labor scarcity, followed by computer science (+0.7%) and business (+0.6%).

The impact of augmentation is strongly positive and relatively uniform across training paths—an increase of roughly +10%. Here the role of labor reallocation is less important, because augmentation boosts productivity in all training paths, and the expected augmentation shock is relatively similar across paths. The uncertainty over augmentation has a negligible effect on wages, as the uncertainty is also relatively similar across training paths.

The final result is that, if workers are highly responsive to AI-induced changes, wages increase nearly uniformly across all training paths. The only exception is legal professions, which see a higher increase in their wages (+17%) due to the endogenous risk premium. This uniformity is the outcome that depends most on the responsiveness of training choices, as discussed in Section 5.3.

5.3 The Role of the Responsiveness of Training Choices

The size of the distributional effects above depends on how responsive training choices are to differences in wage premia. Figures 6 and 7 compare the net effects under the three scenarios of Section 4: high ($\zeta = 0.053$), medium ($\zeta = 0.5$), and low ($\zeta = 1.2$) responsiveness. Since the three scenarios are equally plausible, the differences across them measure how much of each result rests on this parameter.

Three results hold in all scenarios. First, the direction of labor reallocation and the ranking of training paths are the same: legal professions, business, and computer science lose workers, while education, technical manual jobs, and no training gain them. Second, AI raises average expected wages by about +15% to +16% in all three scenarios. Third, the relative importance of uncertainty in the reallocation is the same: uncertainty accounts for about a fifth of the automation-driven outflow from legal professions, and for about a tenth of that from computer science and business, whatever the responsiveness of training choices. This is because, in general equilibrium, the response of employment to any shock to the certainty equivalents, whether it comes from expected shocks or from uncertainty, is scaled by the same factor $1/(\zeta + \beta/\epsilon)$, as in the proof of Proposition 2.

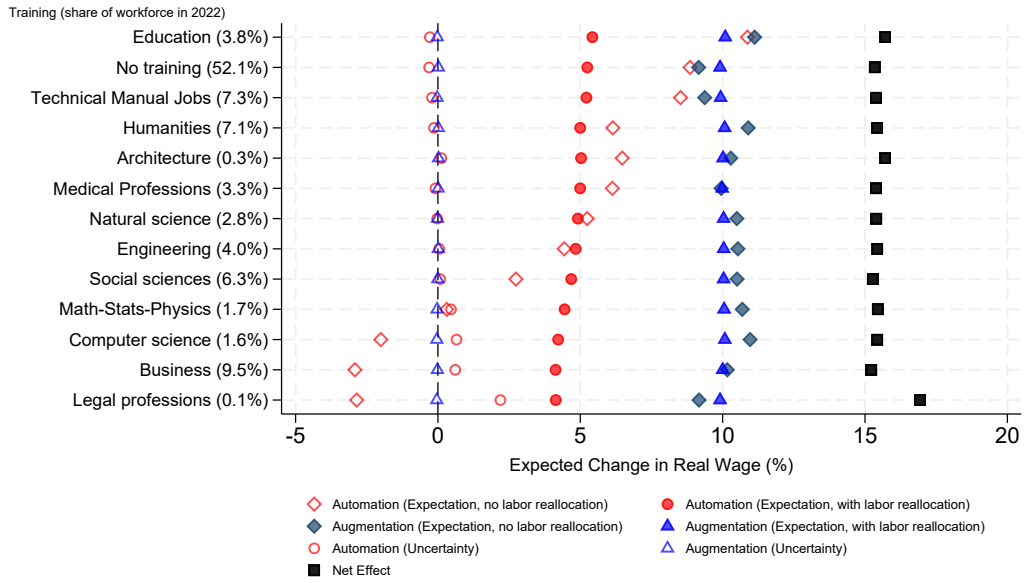


Figure 5: Decomposition of AI-induced expected wage changes across training paths.

Notes: This figure decomposes, in the benchmark (high-responsiveness, $\zeta = 0.053$) scenario, the percentage change in the expected real wage into six components: expected automation with no labor reallocation (red hollow diamonds), expected automation with labor reallocation (red solid circles), uncertainty over automation (red hollow circles), expected augmentation with no labor reallocation (navy solid diamonds), expected augmentation with labor reallocation (blue solid triangles), and uncertainty over augmentation (blue hollow triangles). The total net wage change is shown as black squares. See Section 4 for details on the benchmark calibration. See Section 3 for details on the expected AI shocks and their uncertainty.

Two results, instead, depend on the responsiveness of training choices. First, the magnitude of labor reallocation: legal professions lose 16%, 9%, and 6% of their workforce under high, medium, and low responsiveness, respectively, and education gains 8%, 5%, and 3%. The endogenous risk premium scales accordingly, from +2.2% to +1.2% and +0.7% in legal professions. Second, the distribution of wage gains across training paths. When training choices are highly responsive, workers move until the scarcity in exposed paths and the congestion in safer paths offset the direct impact of AI, and wage gains are nearly uniform (between +15% and +17%). When they are less responsive, fewer workers leave the exposed paths, the scarcity channel is weaker, and the adjustment shifts from employment to wages: under medium responsiveness, wage gains range from about +12% in legal professions, business, and computer science to about +19% in education, and under low responsiveness from about +10% to about +21%. In other words, the responsiveness of training choices determines whether AI mainly reallocates workers across training paths or opens wage gaps between them.

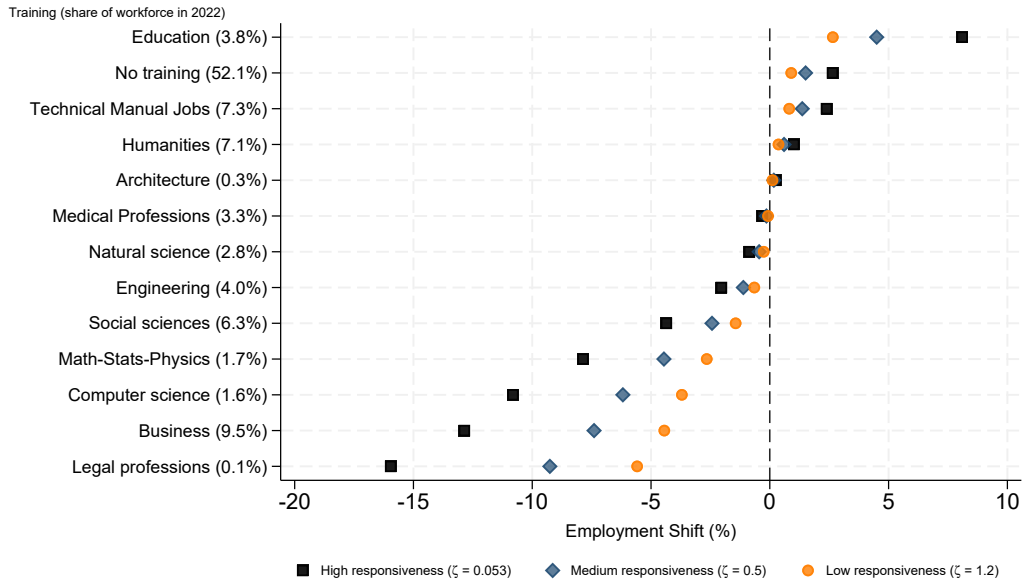


Figure 6: Net AI-induced employment shifts across training paths under alternative responsiveness of training choices.

Notes: This figure reports the net percentage change in employment shares (relative to the pre-AI 2022 baseline) under high ($\zeta = 0.053$, benchmark; black squares), medium ($\zeta = 0.5$; navy diamonds), and low ($\zeta = 1.2$; orange circles) responsiveness of training choices. Training paths are ranked by their net employment shift in the benchmark. In each scenario, the training costs $\bar{C}_j^*$ are recalibrated to reproduce the 2022 employment shares, and all other parameters are held at their baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\eta = 5$). See Section 4 for the derivation of the three values of ζ.

6 Sensitivity Analysis

In this section, I test the sensitivity of the main results to alternative parameterizations of the model. I focus on two key parameters: the risk aversion coefficient η and the probabilities of automation and augmentation (p_A and p_γ). Each alternative is compared with the high-responsiveness scenario. The results are qualitatively similar across all alternative specifications, but the magnitude of the effects varies with the parameter values.

As far as the risk aversion coefficient η is concerned, I test a low risk aversion scenario with $\eta = 2$ and a high risk aversion scenario with $\eta = 10$. Figure A1 shows that labor reallocation is largely similar across the three scenarios, with the most exposed paths (legal professions, business, and computer science) losing up to 20% of their workforce, while the safer paths (technical manual jobs, education, and no training) gain up to +9%. The main difference is that higher risk aversion amplifies the flight from uncertainty, leading to larger employment shifts in the most exposed paths. As a result, the endogenous risk premium is also larger in the high risk aversion scenario,

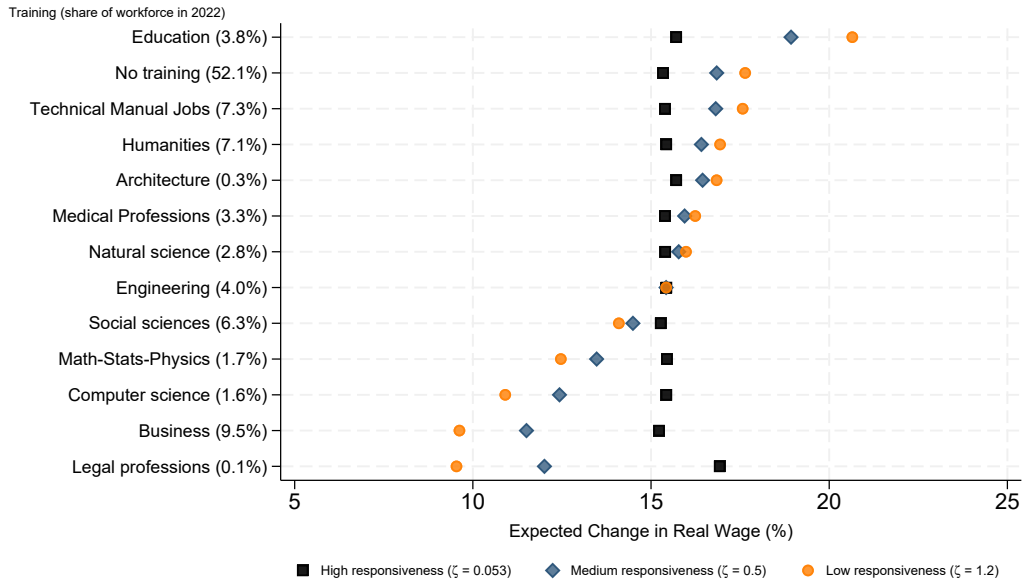


Figure 7: Net AI-induced expected wage changes across training paths under alternative responsiveness of training choices.

Notes: This figure reports the net percentage change in the expected real wage under high ($\zeta = 0.053$, benchmark; black squares), medium ($\zeta = 0.5$; navy diamonds), and low ($\zeta = 1.2$; orange circles) responsiveness of training choices. Training paths are ranked by their net employment shift in the benchmark, as in Figure 6. In each scenario, the training costs $\bar{C}_j^*$ are recalibrated to reproduce the 2022 employment shares, and all other parameters are held at their baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\eta = 5$). See Section 4 for the derivation of the three values of ζ.

with expected wages in legal professions rising by approximately 20% in the high risk aversion scenario, compared to 17% in the baseline.

Figure A3 and Figure A4 show the results of the alternative scenario where the probabilities of automation and augmentation are set to $p_A = 0.8$ and $p_\gamma = 0.8$. The results are qualitatively similar to the baseline scenario, but the labor reallocation is stronger, since workers believe that AI is more likely to be eventually successful. As a result, training paths that are more exposed to automation lose up to 24% of their workforce (compared to 16% in the baseline), while the safer paths (technical manual jobs, education, and no training) gain up to +11% (compared to +8% in the baseline). The wage impact is also stronger, with wages rising by roughly +26% in all training paths (compared to +15% in the baseline).

Overall, the main implications of the model are robust to alternative parameterizations, confirming the importance of accounting for the labor supply response to AI. In all specifications, the qualitative results are the same: the most exposed paths lose a significant share of their workforce, while the safer paths (technical manual jobs, education, and no training) gain a significant number of workers. This labor reallocation

generates a general equilibrium scarcity effect that mitigates the negative wage impact of AI in the most exposed paths, while compressing wages in the safer paths.

7 Conclusion

This paper studies how Artificial Intelligence reshapes human capital investment when its impact is uncertain and heterogeneous across training paths. I develop a task-based model in which risk-averse agents make irreversible training choices to maximize a certainty equivalent that penalizes AI-related wage variance, and I calibrate the framework to the US labor market using ACS and BLS data together with task-level automation and augmentation exposure from Althoff and Reichardt (2026) and Massenkoff and McCrory (2026).

The model delivers three main results. First, expected automation is the dominant driver of labor reallocation: workers systematically abandon highly exposed cognitive paths—legal professions, business, and computer science—and crowd into technical manual jobs, education, and the no-training option. The direction of this reallocation is the same in all scenarios, while its size depends on the responsiveness of training choices. Expected augmentation, by contrast, plays a minor role in reallocation because its expected impact and uncertainty are relatively similar across training paths.

Second, AI raises average expected wages by about +15% to +16% in all scenarios, but the endogenous reallocation of labor shapes how these gains are distributed. The scarcity of workers in highly exposed paths pushes up the price of what they produce and their wages, while the congestion in safer paths compresses wages there. When training choices are highly responsive, these two forces nearly offset the direct impact of automation, yielding a uniform wage increase of about +15%. When they are less responsive, the scarcity channel is weaker, and AI opens wage gaps across training paths, with gains ranging from about +10% in the most exposed paths to about +21% in the safest one.

Third, automation uncertainty generates an endogenous risk premium, a compensating differential for AI-related risk, which ranges from +0.7% to +2.2% depending on the responsiveness of training choices. Risk-averse individuals flee volatile training paths independently of their expected returns, creating additional labor scarcity that raises expected wages in those paths above what the expected productivity effects alone would predict.

Taken together, these results highlight the importance of accounting for the labor supply response to AI. A training path facing severe automation risk may nonetheless see the scarcity channel mitigating the negative wage impact, while a path perceived as safe may experience disappointing wage growth due to labor market congestion. How

strong these forces are depends on how responsive training choices are to AI, which makes better evidence on this responsiveness a priority for assessing the distributional consequences of AI.

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A Sensitivity Analysis Figures

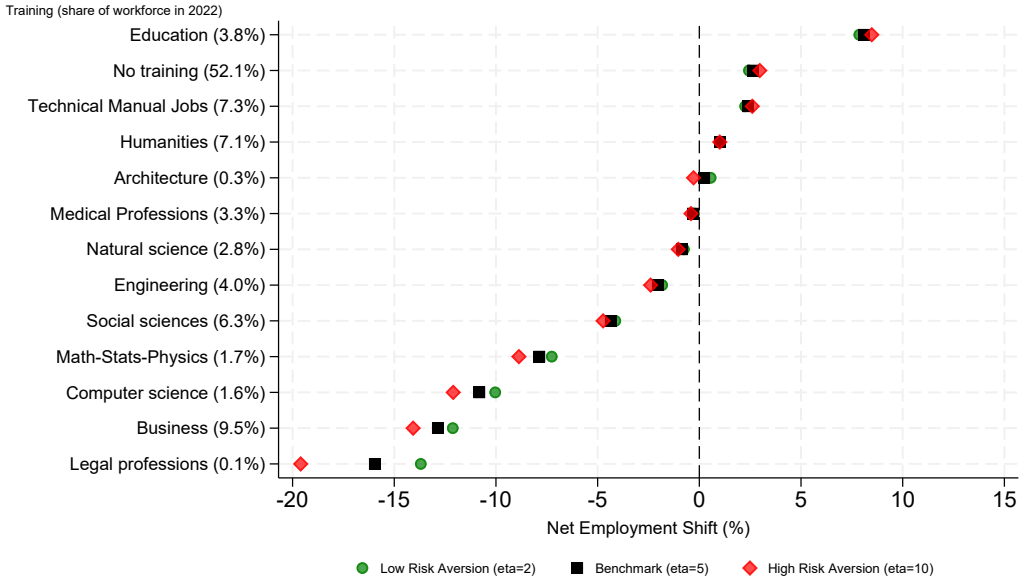


Figure A1: Employment Reallocation across Training Paths: Baseline vs. Alternative Risk Aversion Scenarios.

Notes: This figure compares the net percentage change in employment shares (relative to the pre-AI 2022 baseline) across training paths under three risk aversion scenarios: low risk aversion ($\eta = 2$, green circles), baseline ($\eta = 5$, black squares), and high risk aversion ($\eta = 10$, red diamonds). All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\zeta = 0.053$). See Section 4 for details on the baseline calibration.

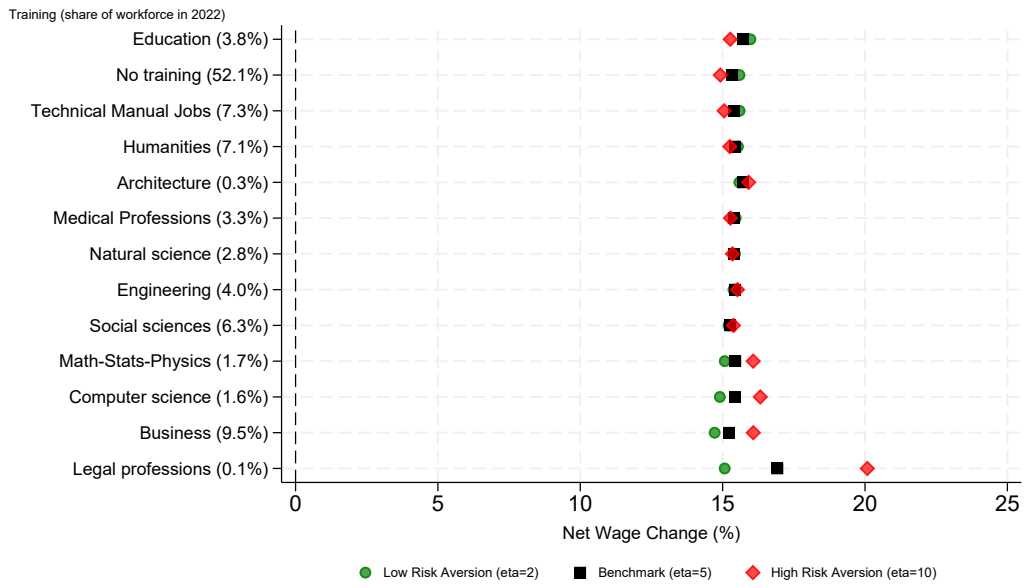


Figure A2: Wage Impact across Training Paths: Baseline vs. Alternative Risk Aversion Scenarios.

Notes: This figure compares the net percentage change in the expected real wage (relative to the pre-AI 2022 baseline) across training paths under three risk aversion scenarios: low risk aversion ($\eta = 2$, green circles), baseline ($\eta = 5$, black squares), and high risk aversion ($\eta = 10$, red diamonds). All other parameters are held at baseline values ($p_A = p_\gamma = 0.5$, $\kappa = 0.5$, $\zeta = 0.053$). See Section 4 for details on the baseline calibration.

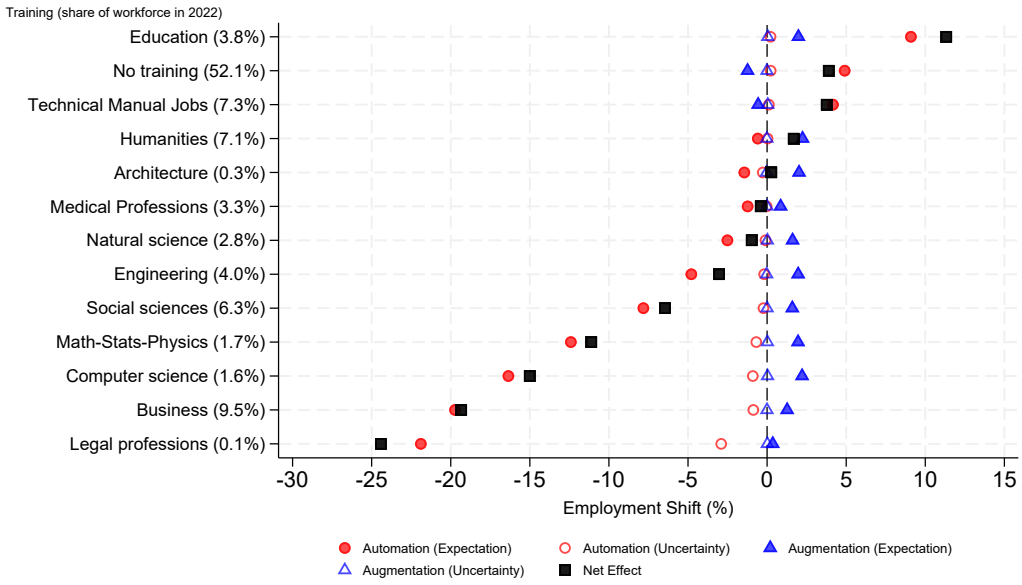


Figure A3: Employment Reallocation across Training Paths: Alternative Automation/Augmentation Probabilities ($p_A = p_\gamma = 0.8$).

Notes: This figure decomposes the percentage change in employment shares (relative to the pre-AI 2022 baseline) into four channels: expected automation (red circles), uncertainty over automation (red hollow circles), expected augmentation (blue triangles), and uncertainty over augmentation (blue hollow triangles). The total net effect is shown as black squares. This specification raises the adoption probabilities to $p_A = p_\gamma = 0.8$ (agents believe there is an 80% chance that theoretically possible automation or augmentation will materialize), with all other parameters at baseline values ($\eta = 5$, $\kappa = 0.5$, $\zeta = 0.053$). See Section 3 for details on how the AI shocks are constructed.

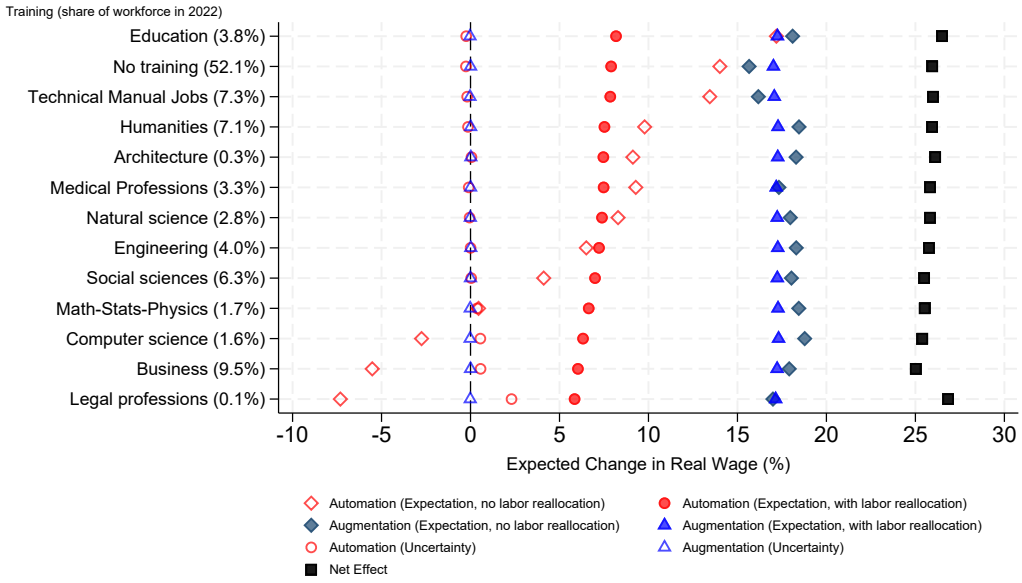


Figure A4: Wage Impact across Training Paths: Alternative Automation/Augmentation Probabilities ($p_A = p_\gamma = 0.8$).

Notes: This figure decomposes the percentage change in the expected real wage into six components: expected automation with no labor reallocation (red hollow diamonds), expected automation with labor reallocation (red solid circles), expected augmentation with no labor reallocation (navy solid diamonds), expected augmentation with labor reallocation (blue solid triangles), uncertainty over automation (red hollow circles), and uncertainty over augmentation (blue hollow triangles). The total net wage change is shown as black squares. This specification uses $p_A = p_\gamma = 0.8$, with all other parameters at baseline values ($\eta = 5$, $\kappa = 0.5$, $\zeta = 0.053$). See Section 3 for details on how the AI shocks are constructed.